SKYG

SKYG SPEED 90B

USER MANUAL

REVISION C6B-20 · 2026-10-16



PACKAGE CONTENTS

Manual

SkyG Speed 90B user manual (this document).

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

STEP

Every printed part and the full assembly, for editing in any CAD program.

Wiring

The full-size wiring diagram and the whole electrical integration (cut list, pinouts, routing, tests) are chapters of this manual: no separate files.

ArduPilot

Setup document and a starting .param file for the SpeedyBee F405 WING MINI.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

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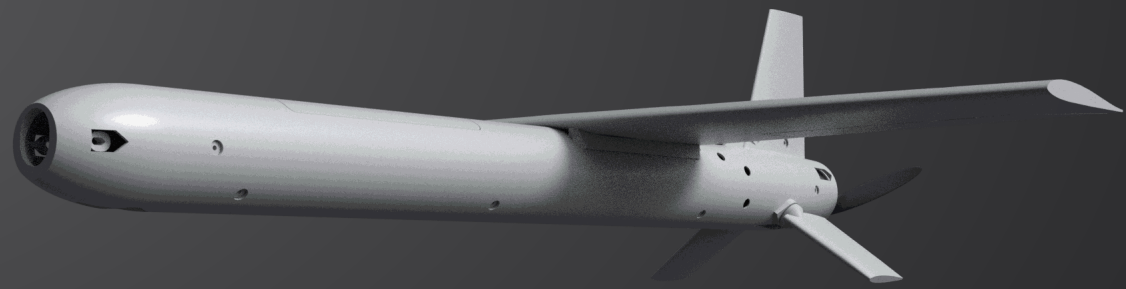
Flight test and speed stages 69

SKYG SPEED 90B

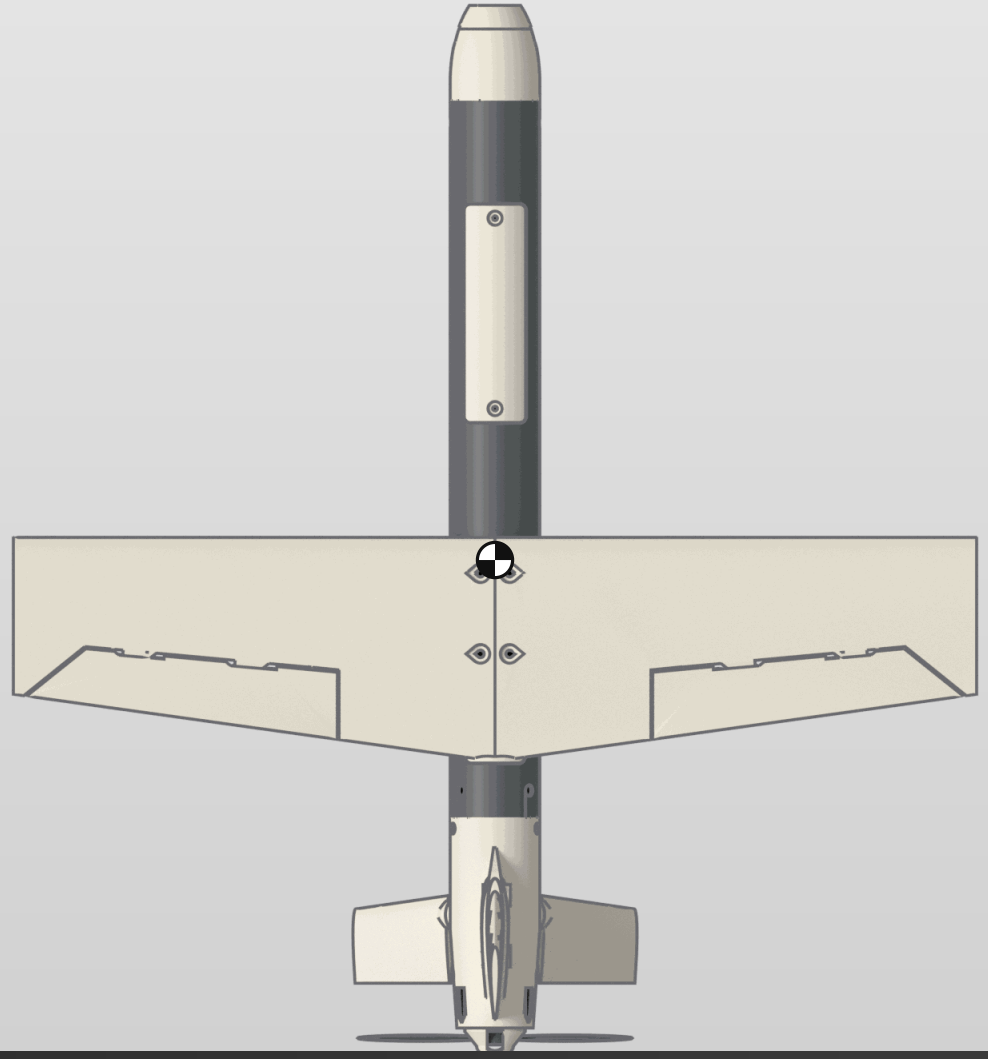
SkyG Speed 90B is a compact, hand-launched speed aircraft: a 40 × 38 mm carbon-tube fuselage (320 mm, cut to a 1:1 template) carrying a printed equipment tray, with a printed camera nose plugged into its front and a printed tail module on its aft end, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

Every part is printed in solid (not foaming) filament on a 0.4 mm hardened-steel nozzle, with no glass skin: PETG-CF (wing halves, ailerons, tail panels, motor plate, spinner, equipment tray, tail coupler, wing-nut bars, hatch lips, battery front stop, wing-window filler, countersunk washers and servo horns), PETG (camera noses, VTX shelf, lip shims, hatch cover and GPS carrier) and ASA (DJI duct insert, fin and tail module). Steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Three interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera and a DJI O4 Air Unit (behind a 1.0 mm clear window).

It is built for straight-line speed: about 77 m/s as printed and 78 m/s finished in the model, flown up through never-exceed stages to 90 m/s.



GENERAL AIRCRAFT DATA



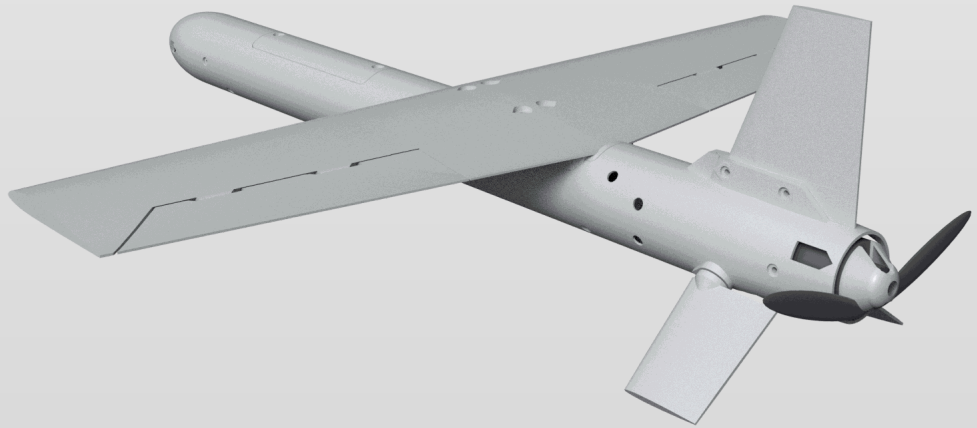
Wingspan	430 mm
Wing area	3.65 dm²
Length	467 mm
Centre of gravity	10 mm behind the wing leading edge at the root
All-up weight	530 g (micro-camera nose)
Stall speed	13.0 m/s
Top speed (model)	77–78 m/s
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	104 g/dm²

AERODYNAMIC DESIGN

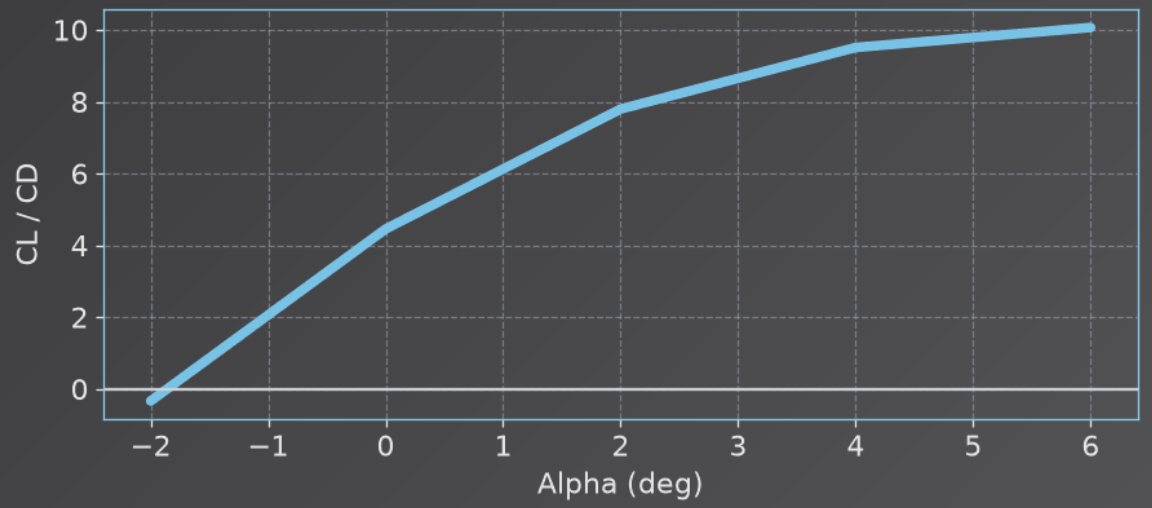
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 12 % of the mean chord, 7.0 % static margin ahead of the neutral point (x 261 mm from the nose datum; accepted band 6–8 %): stable without a tail download penalty at speed.

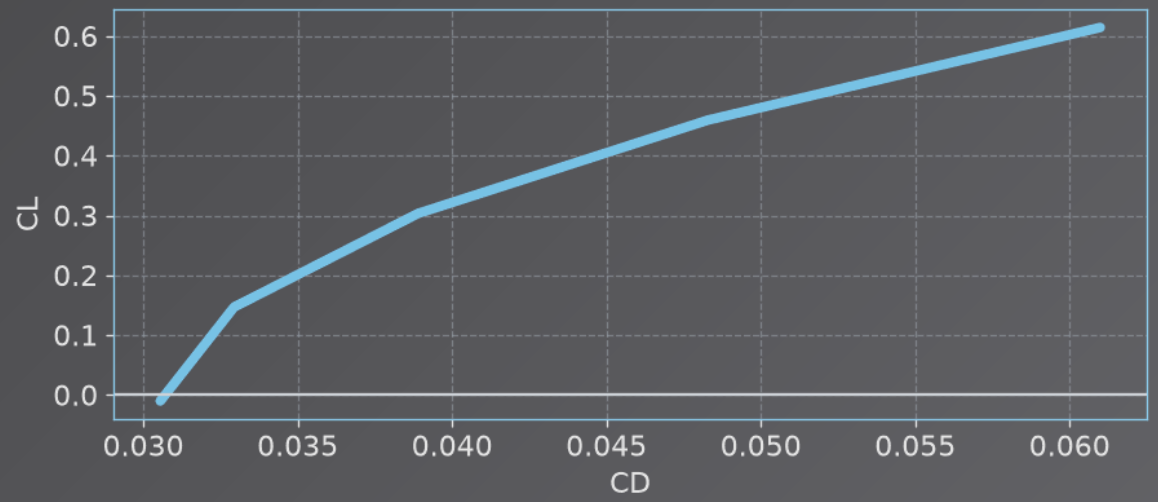
Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

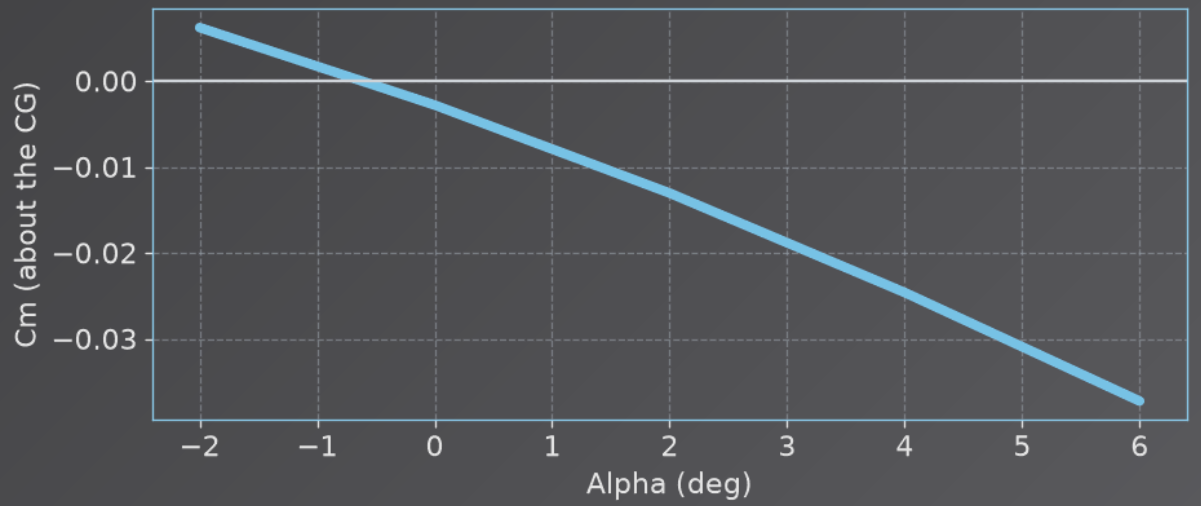
Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 116 m/s, above 1.2 × the 90 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.

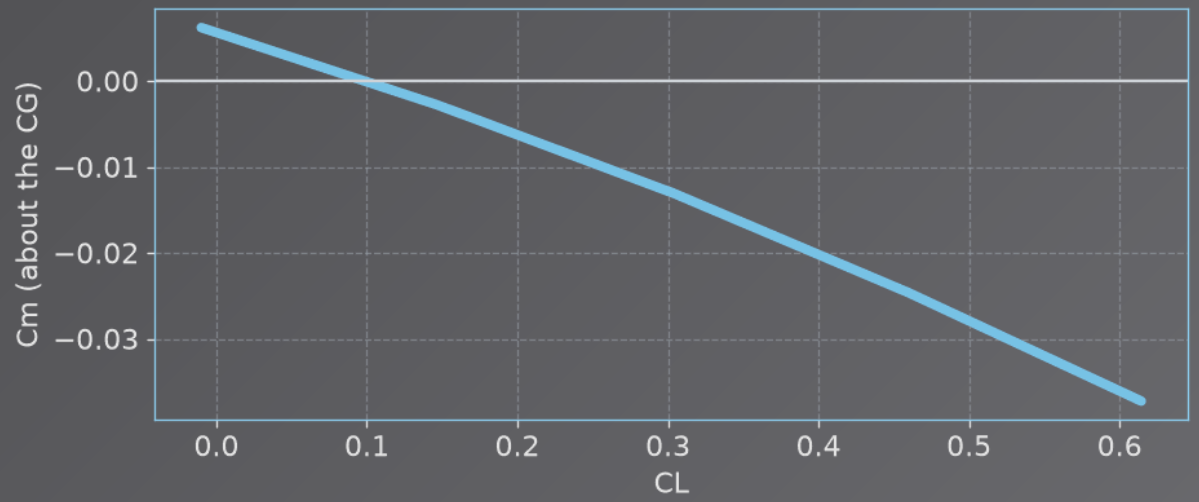


AERODYNAMIC CHARACTERISTICS









Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 10.1 at 6.0°; Cm about the design CG.

PERFORMANCE AND SPEED STAGES

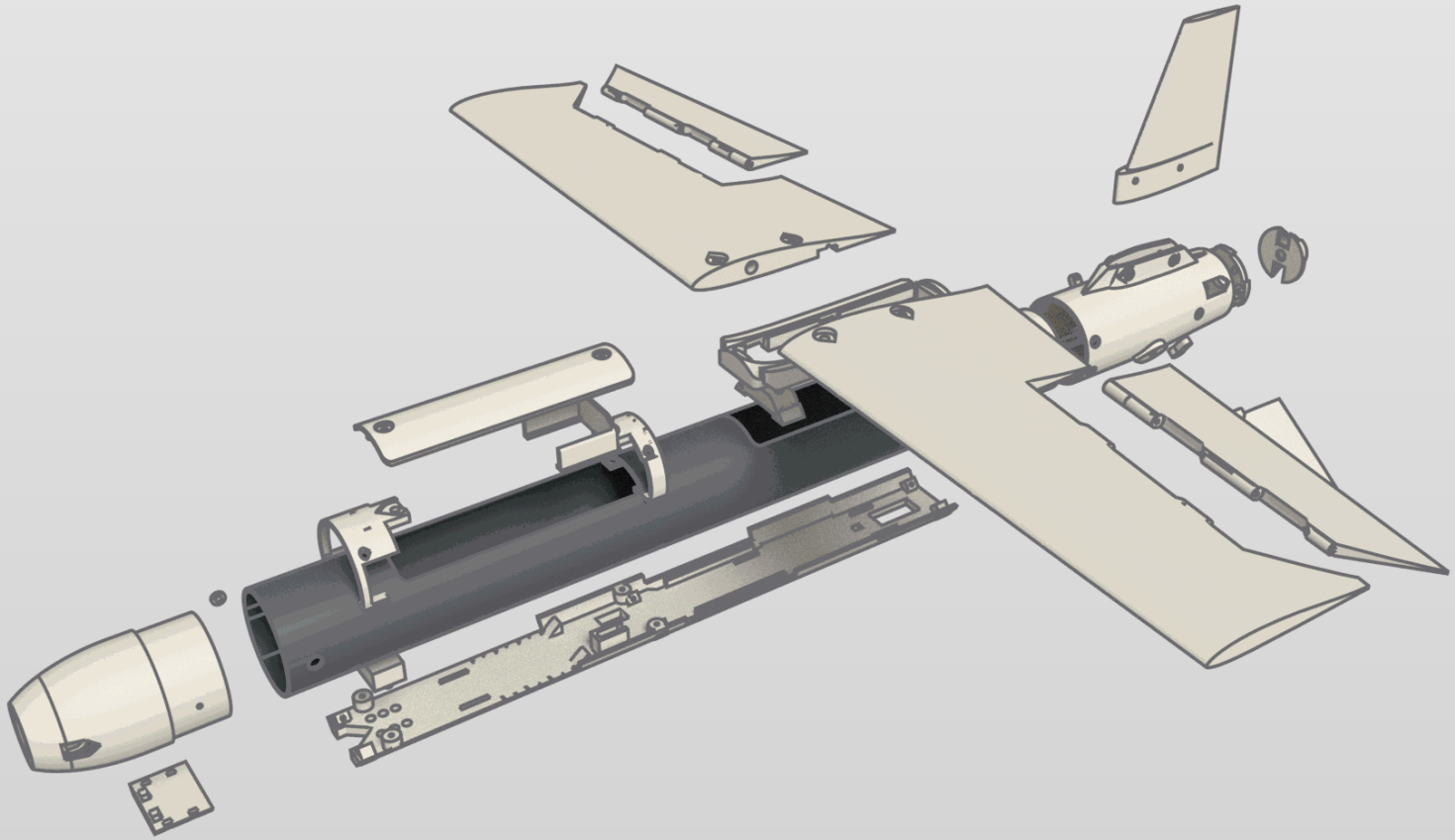
Motor	T-Motor F50 2207 2150KV
Propeller	Gemfan 5x7.5E Vortex
Battery	Tattu R-Line 4S 650 75C
Static thrust	870 g at 36 A (thrust / weight 1.64)
Throttle cap (prop rpm rating)	100 % (none needed)
Top speed	77 m/s as printed, 78 m/s finished
Full-throttle time	about 0.9 min per pack (bursts of ≤ 10 s)

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	33 %	24 m/s
35 m/s	37 %	28 m/s
40 m/s	42 %	32 m/s
45 m/s	47 %	37 m/s
50 m/s	53 %	42 m/s
55 m/s	58 %	46 m/s
60 m/s	63 %	50 m/s
65 m/s	68 %	53 m/s
70 m/s	74 %	58 m/s
75 m/s	79 %	62 m/s
80 m/s	85 %	66 m/s
85 m/s	91 %	71 m/s
90 m/s	96 %	74 m/s

No airspeed sensor: each stage is held by a throttle limit on a transmitter switch (LAUNCH = full cap for the throw and climb-out only, STAGE = the limit). Level runs only; speed from GPS two-way passes.

FILES

Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons, tail panels and servo horns are separate files because they are mirror images. Print from the supplied Bambu Studio projects (.3mf): they carry each part's orientation and settings. The interactive viewer in the package shows every part, purchased item and cable in place.



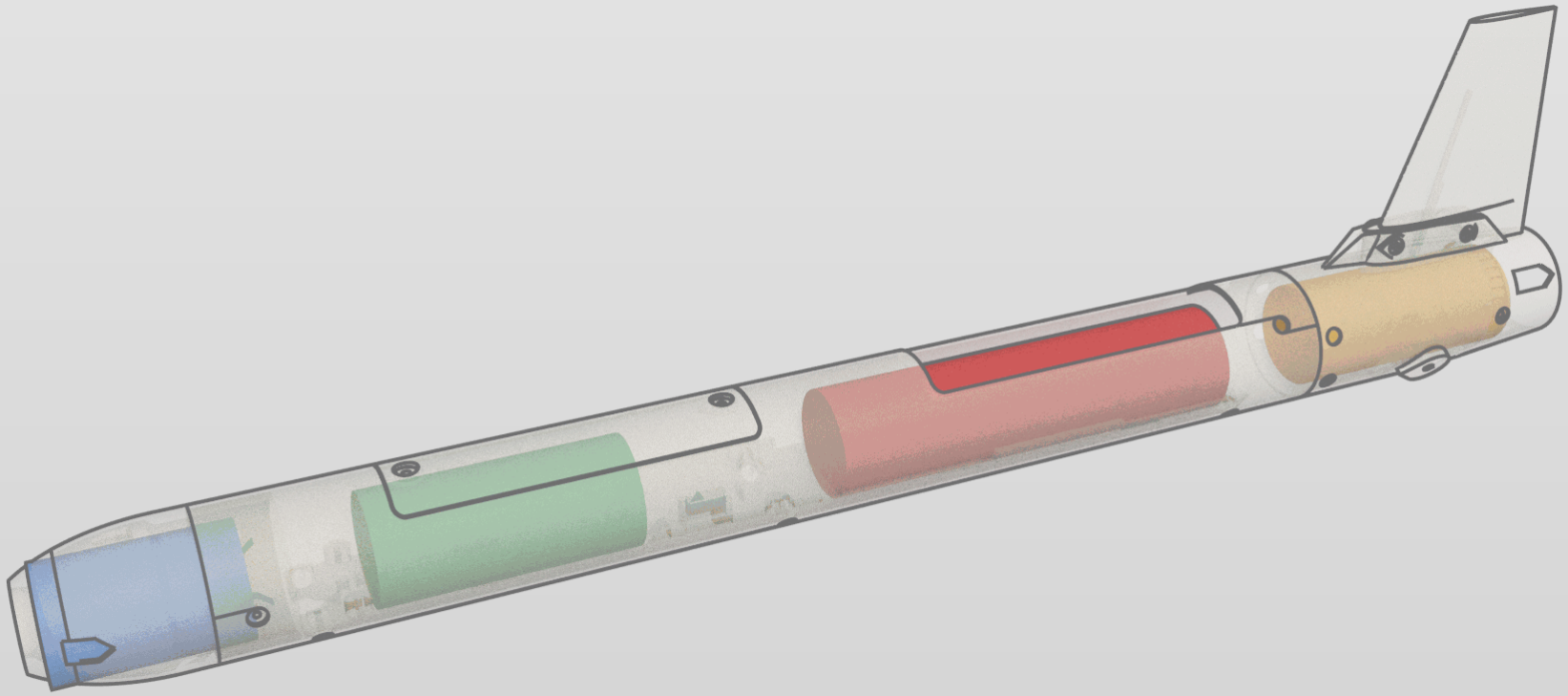
COMPARTMENT LAYOUT

BATTERY BAY
Tattu R-Line 4S 650 75C, 60 × 31 × 21 mm max; pack centre x 120–130 (rear edge x 150–160, the ticks on the floor edges); rear stop at x 160 (the pack's rear edge, GPS behind)

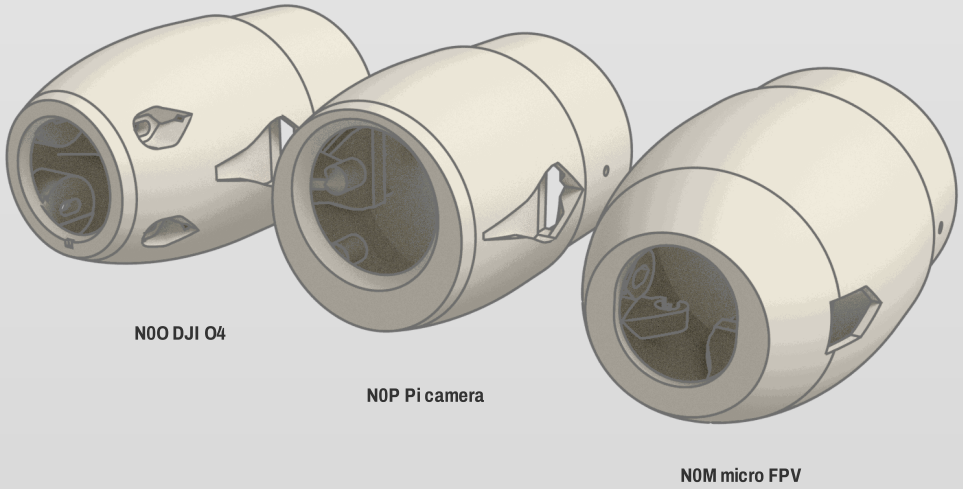
WING BAY
ESC flat at the bottom, receiver on the ESC, flight-controller stack behind it, aileron servo aft; all on the equipment tray, which slides out of the tube's front

TAIL MODULE
Two tail servos, pushrods, horns and the motor, the fin's receiver antenna; plugs off at the MR30, the antenna's u.FL joint and the servo plugs

NOSE BAY
Camera and VTX (or the DJI O4 unit) in the removable nose

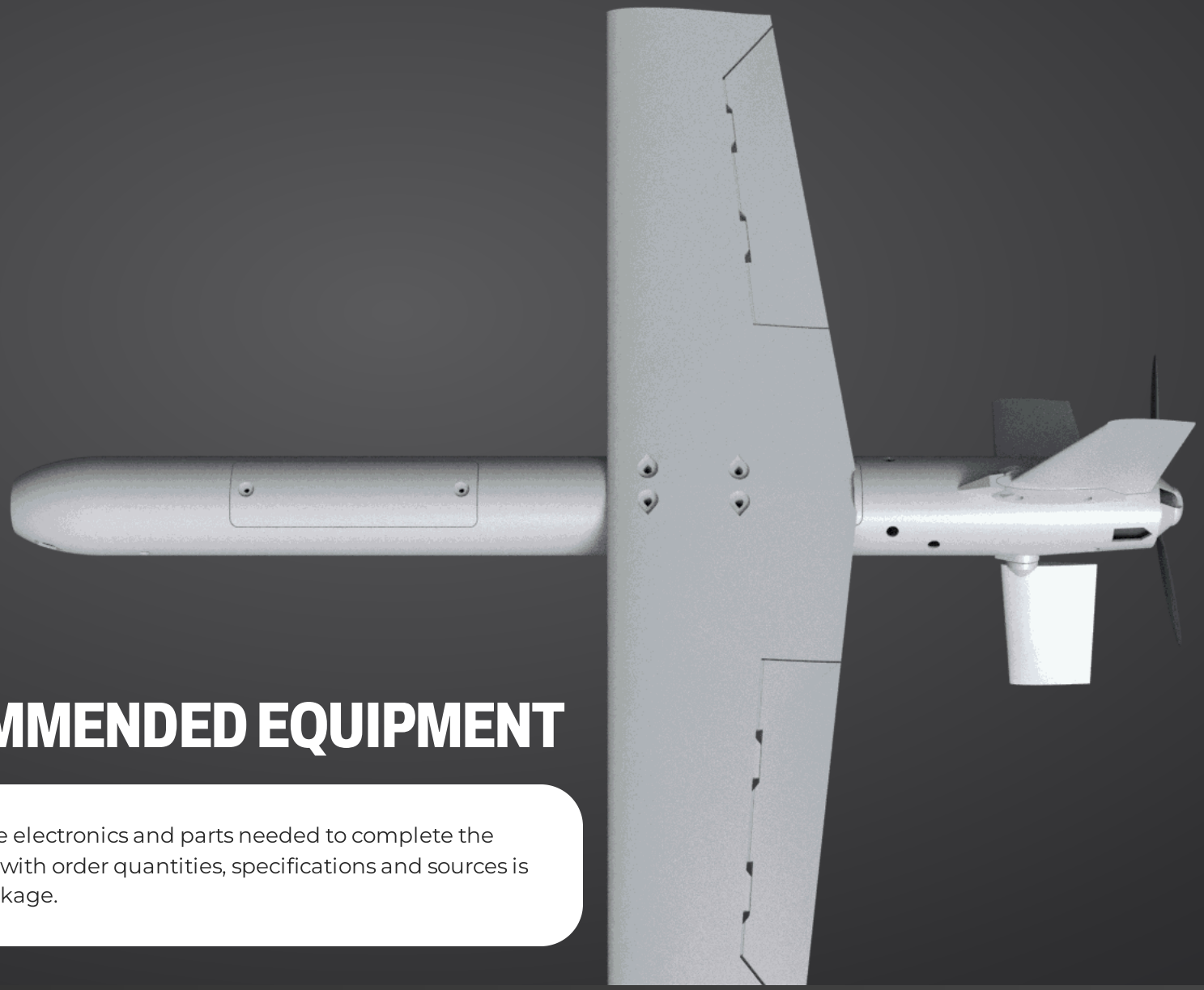


CAMERA NOSE VARIANTS



Nose	Camera	Battery pack position for balance	Static margin
N0M	19 mm micro FPV camera + 20 × 20 VTX	centre x 128, rear-edge tick 158	7.0 %
N0P	Raspberry Pi Camera Module 3 (test option)	centre x 120 (forward end), rear-edge tick 150	3.3 % (tail-heavy: add nose ballast); 6.7 g nose ballast gives 7 %
N0O	DJI O4 Air Unit (Lite), ducted, behind a 20 × 1.0 mm clear window	centre x 130 (rear stop), rear-edge tick 160	8.3 % (nose-heavy, above the band)

Pack position = its centre, x in mm from the nose tip; the floor ticks mark the pack's rear edge (centre + 30 mm). Accepted static margin 6–8 %.



RECOMMENDED EQUIPMENT

This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT	MODEL / SPECIFICATION
Motor	T-Motor F50 2207 2150KV
Propeller (fixed)	Gemfan 5x7.5E Vortex
ESC	HDZero Halo 4in1 70A ESC AM32
Flight controller	SpeedyBee F405 WING MINI
GPS + compass	TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos	3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery	Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g listed
Receiver	RadioMaster RP1 V2
FPV camera	Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna	20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)	DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
Prop adapter ring	1
XT30 connector pairs	1
Carbon tube	1
Music wire 2.0 mm	2
Pushrods 1.2 mm with Z-bend	4
Pushrod keepers for 1.2 mm wire	4
Steel countersunk screw M3 x 16	2
M3 hex nut	4
M2 x 6 countersunk screws (joints)	4
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	1
Heat shrink + silicone wire	as needed
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 100 mm	2
JST-SH/GH lead with plug, 4-6 pin	1
Camera pivot screws	2
Motor screws M3	4
Thin M5 nut	1
Steel countersunk screw M3 x 12	2
Steel cup washer M3 (countersunk finishing washer)	4

ITEM	QTY
Steel cup washer M2 (countersunk finishing washer)	2
M2 x 6 countersunk screws (motor plate)	2
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
M2 x 5 pan head (O4 camera mount) (DJI nose)	4
Foam strip + tape (O4 module) (DJI nose)	1
O4 3-in-1 lead extension (DJI nose)	1
Clear window (DJI nose)	1
Vent tape	1
15T servo arms	1
u.FL extension 1.13 mm, about 250 mm	1
Dupont housings + crimps (1-pin, 4-pin)	3
Carbon-fibre tube 40 x 38 mm	320 mm
M2 x 6 countersunk screws (tube)	10
Structural epoxy	~3 g
Thin CA	~1 g
M2 x 6 countersunk screws (hatch)	2
M2 x 6 socket head screws (hatch lips)	4
Brass M2 heat-set inserts	4
M2 x 12 countersunk screws (fin)	2
USB-C extension with socket	1
Conformal coating	~2 ml

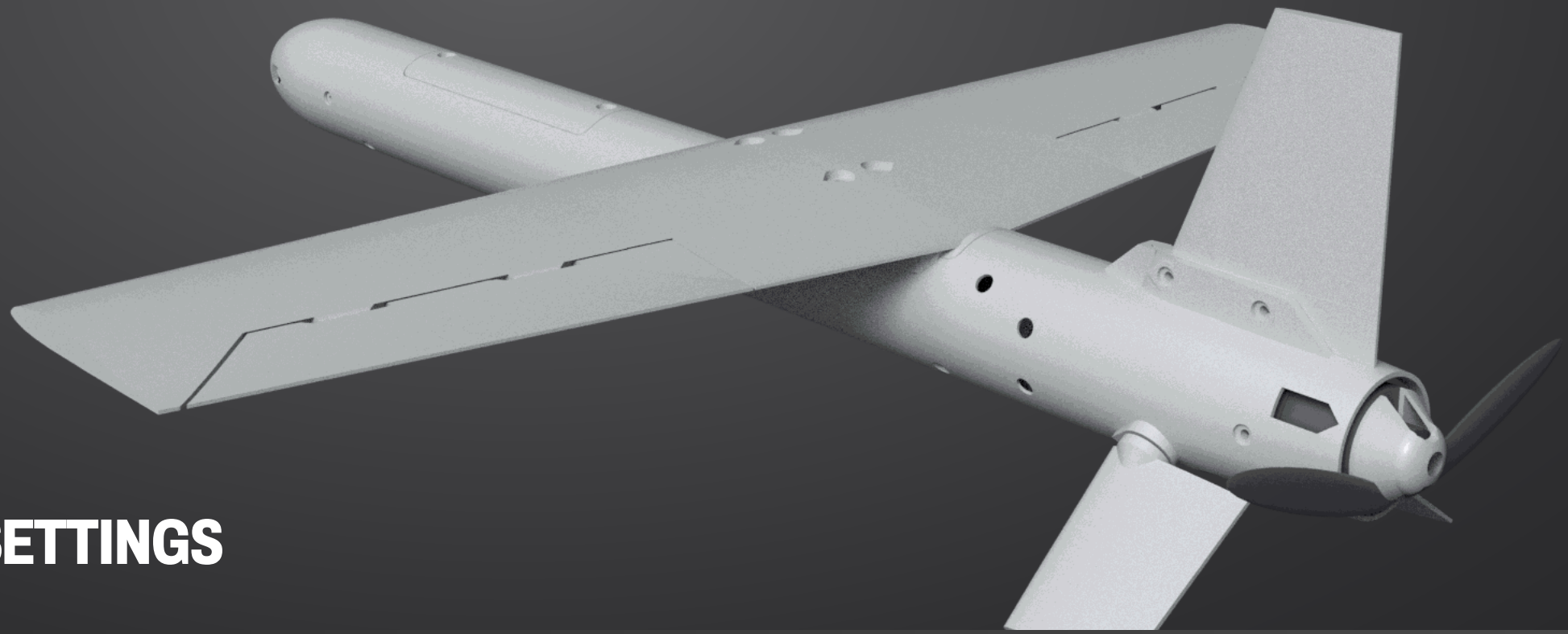
Not counted above. Tools and test equipment: Transmitter; Balance charger; Z-bend pliers, 1.1 mm drill, 1.2 mm drill (as a hand reamer), small flat file; Thrust stand / wattmeter; Diamond cut-off discs + dust protection. Pi-camera nose option: Raspberry Pi camera; CSI camera ribbon; Pi board screws; Nose ballast; Clear window (Pi nose). Filament: see the parts list.

POWERTRAIN

Motor and prop. T-Motor F50 2207 2150KV (6S-rated, 24 V; Ø27 x 29.5 mm incl. shaft, 28.56 g, 16 x 16 mm M3 with 4 mm thread depth, 5 mm shaft, M5 x 10.75 mm prop thread, 50 mOhm, I0 1.6 A @10 V; 54.8 A / 1273 W peak): the SkyG Speed 60 motor, same mount. Prop: Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre polyamide, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring; Robocombo stock). Gemfan publishes no rpm rating: each batch is spin-tested to 45,600 rpm (1.2 x the 38,000 rpm governor ceiling) behind a shield before flight. Pusher: the 5x7.5R, or the E mounted front face toward the motor with the motor reversed in the ESC; no governor or throttle cap needed on 4S: an unloaded prop on a full pack reaches about 36,100 rpm, under the 38,000 rpm ceiling.

No throttle cap needed. The prop's operating ceiling is 38,000 rpm. On a full 4S pack the 2150 KV motor spins it to at most 36,120 rpm unloaded, so full throttle is safe (THR_MAX 100 %). Never fit a higher-KV motor or a bigger pack without recalculating this; spin-test each prop batch and inspect the blades after every thrust-stand run and flying day.

Battery. Tattu R-Line 4S 650 75C (60 x 31 x 21 mm, 74 g listed; the balance uses 78 g: weigh yours); XT30. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 870 g at 36 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS

Filament per part (no glass skin): **PETG-CF** wing halves, ailerons, tail panels, motor plate, spinner, equipment tray, tail coupler, wing-nut bars, hatch lips, battery front stop, wing-window filler, countersunk washers and servo horns; **PETG** camera noses, VTX shelf, lip shims, hatch cover and GPS carrier (radio windows over the camera, GPS and antennas: never carbon-filled filament there); **ASA** DJI duct insert, fin and tail module (heat: next to the motor and the DJI module; the fin carries the receiver antenna). 0.4 mm hardened-steel nozzle, 0.20 mm layers, wing halves 0.16 mm, ailerons and wing-window filler 0.12 mm; every part fits both the H2S and the H2D build volume (340 × 320 × 325 mm).

GENERAL GUIDELINES (from the supplied print projects)
- Camera noses, DJI duct insert, fin, tail module, tail coupler, wing-nut bars, hatch lips, hatch cover, GPS carrier, battery front stop and countersunk washers: 2 walls, 30 % lightning.
- Motor plate, spinner and servo horns: 4 walls, 40 % grid.
- Wing halves: 2 walls, 8 % gyroid.
- Ailerons: 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid.
- VTX shelf: 2 walls, solid (100 %).
- Equipment tray: 2 walls, 20 % gyroid.
- Lip shims: 1 wall, solid (100 %).
- Wing-window filler: 3 walls, 25 % gyroid.
- Ailerons print together on one plate (AP-ailerons-pair.3mf): each part's layer cools while the other prints.
- Brim: auto brim 5 mm (the slicer adds it where a part needs it); CF filaments need a hardened-steel nozzle, ASA the closed chamber.

SUPPORTS: OFF
Every part is designed to print without supports in the orientation shown, and checked on the slicer's own G-code with these settings: nothing is printed over air and no bridge is longer than 10 mm. Keep supports switched off and use the supplied .3mf files (they carry the settings). The downloads hold ONE set of Bambu Studio projects for the H2S and the H2D: no G-code inside (G-code only fits the printer it was sliced for), tree supports on (auto). Open, select your printer, slice, print. Check the top of the aileron's torque-rod bore on the first aileron. Do not re-arrange or turn the parts on the plate: each one is turned on purpose so the slicer's layer-start seam falls on the trailing edge or a clean line (re-arranging puts it back on corners and holes).

PRINT SETTINGS

FIRST: MEASURE THE TUBE

Measure the carbon tube's bore at both ends before printing anything that fits inside it (the design uses 38.0 mm). If it differs by more than 0.1 mm, the inside parts must be rebuilt for your tube. Print the tail coupler first and try it in both ends of the tube: it must slide in with a light push.

COUPON TESTS

Print the camera nose and try its spigot in the tube's front; slide the tail module onto the coupler; print the tray and slide it the whole length of the tube before cutting any window. Print a 40 mm wing-root section in PETG-CF and check the spar fit.

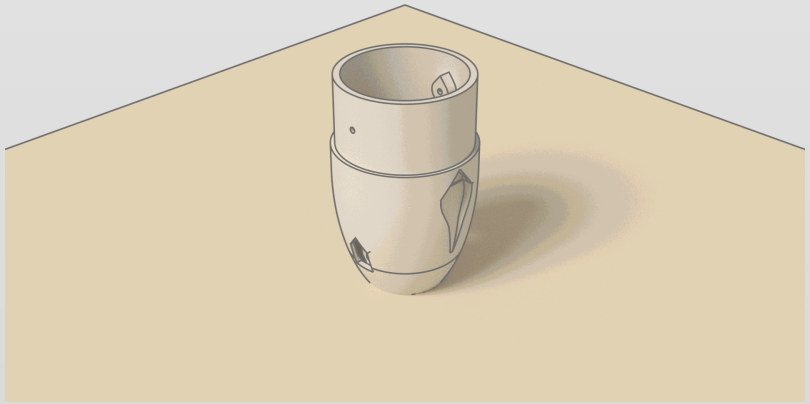
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	PETG	14.0 g
N0P NOSE (PI CAMERA)	PETG	13.4 g
N0O NOSE (DJI O4)	PETG	14.6 g
F3 TAIL MODULE	ASA	16.5 g
HF FRONT HATCH	PETG	4.3 g
WR / WL WING HALVES	PETG-CF	2 × 42.0 g
AR / AL AILERONS	PETG-CF	2 × 5.9 g
SR / SL TAIL PANELS	PETG-CF	2 × 6.0 g
M1 MOTOR PLATE	PETG-CF	2.1 g
FN FIN	ASA	6.8 g
SP SPINNER	PETG-CF	1.5 g
VTX SHELF (N0M)	PETG	1.7 g
HG GPS CARRIER	PETG	1.1 g

PART	MATERIAL	MASS
TR EQUIPMENT TRAY	PETG-CF	25.1 g
BF BATTERY FRONT STOP	PETG-CF	0.9 g
TC TAIL COUPLER	PETG-CF	8.5 g
NB1 WING-NUT BAR FRONT	PETG-CF	1.4 g
NB2 WING-NUT BAR REAR	PETG-CF	1.4 g
HL1 HATCH LIP FRONT	PETG-CF	1.7 g
HL2 HATCH LIP REAR	PETG-CF	1.4 g
LS LIP SHIM	PETG	12 × 0.1 g
CW COUNTERSUNK WASHER (PRINT 18)	PETG-CF	18 × 0.1 g
WF WING-WINDOW FILLER	PETG-CF	8.6 g
DT DJI DUCT INSERT	ASA	4.9 g
AHR / AHL AILERON HORNS	PETG-CF	2 × 0.2 g
THR / THL TAIL HORNS	PETG-CF	2 × 0.3 g

One aircraft with the N0M nose: 254 g printed. Only one camera nose is needed; the others are options.

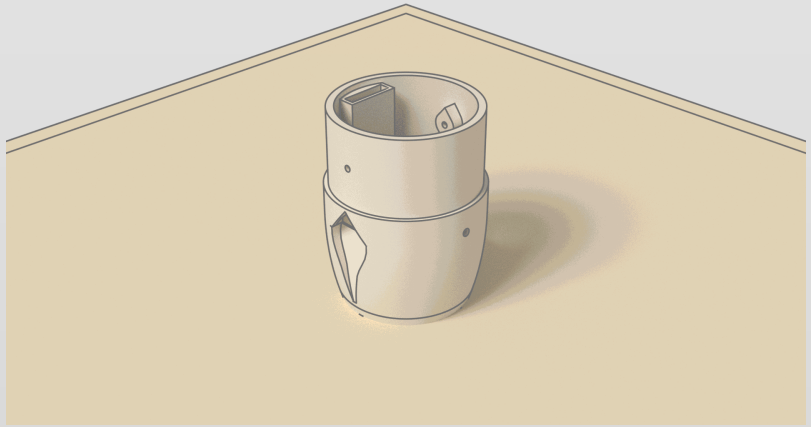
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

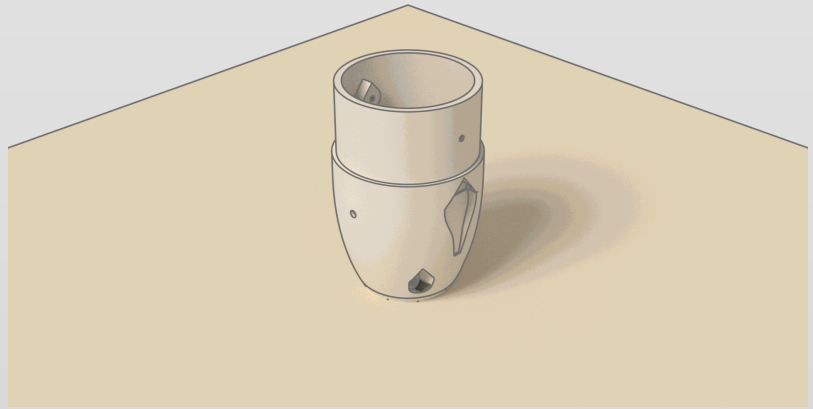
2 WALLS · 30 % LIGHTNING · PETG



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 30 % LIGHTNING · PETG

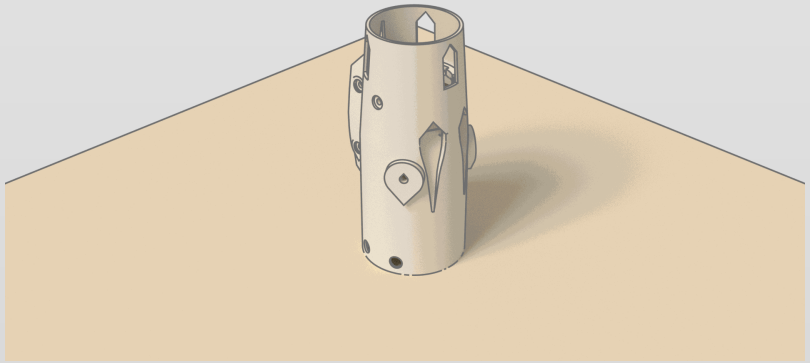


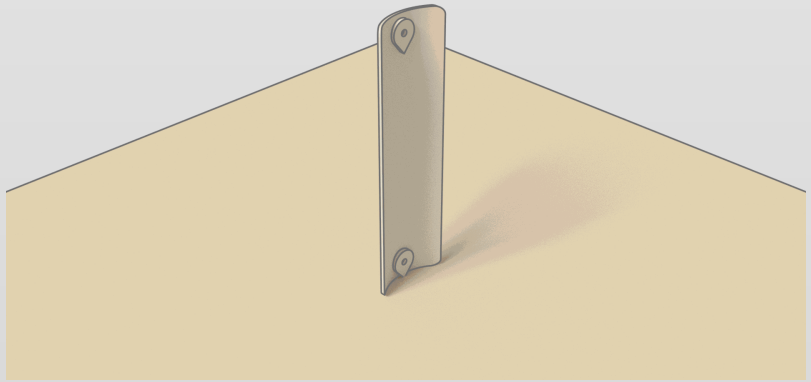
N0O NOSE (DJI O4)

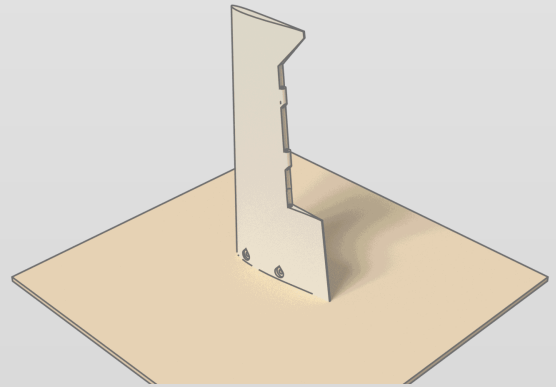
FACE DOWN

2 WALLS · 30 % LIGHTNING · PETG

PARTS ORIENTATION







F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM

2 WALLS · 30 % LIGHTNING · ASA

HF FRONT HATCH

STANDING, FRONT END DOWN

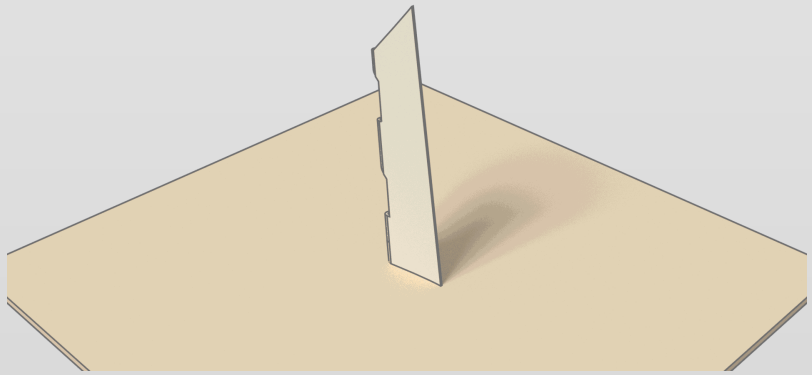
2 WALLS · 30 % LIGHTNING · PETG

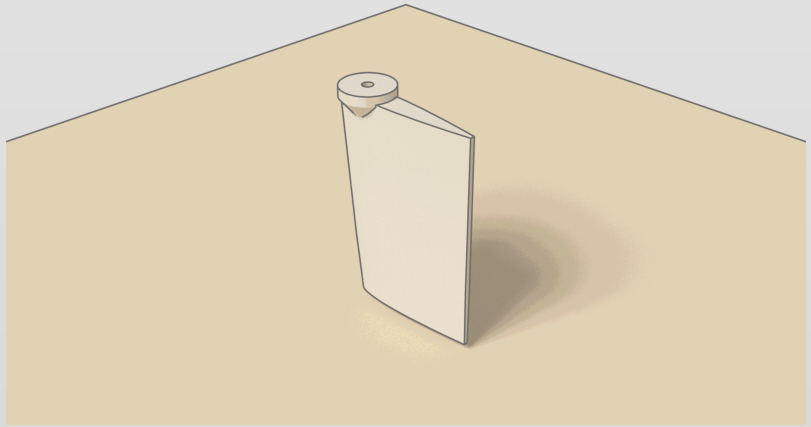
WR / WL WING HALVES

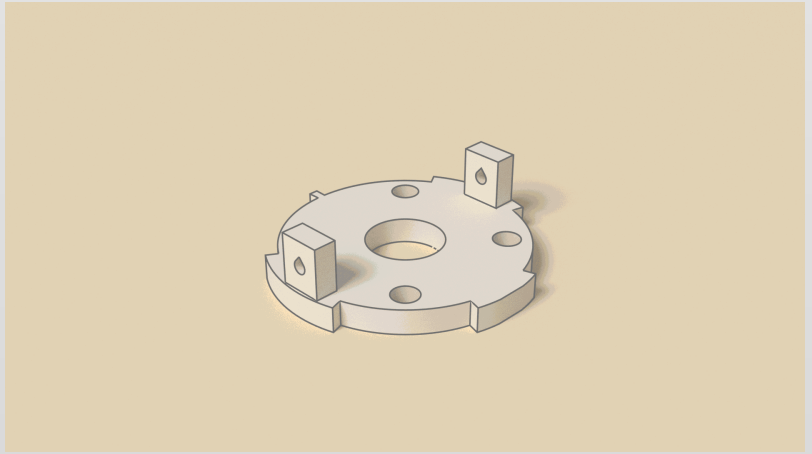
ON THE ROOT RIB

2 WALLS · 8 % GYROID · 0.16 MM LAYERS · PETG-CF

PARTS ORIENTATION







AR / AL AILERONS

ON THE INBOARD END

1 WALL · 5 % GYROID · 0.12 MM LAYERS · PETG-CF

BOTH ON ONE PLATE: AP-AILERONS-PAIR.3MF

SR / SL TAIL PANELS

TIP DOWN, HUB UP

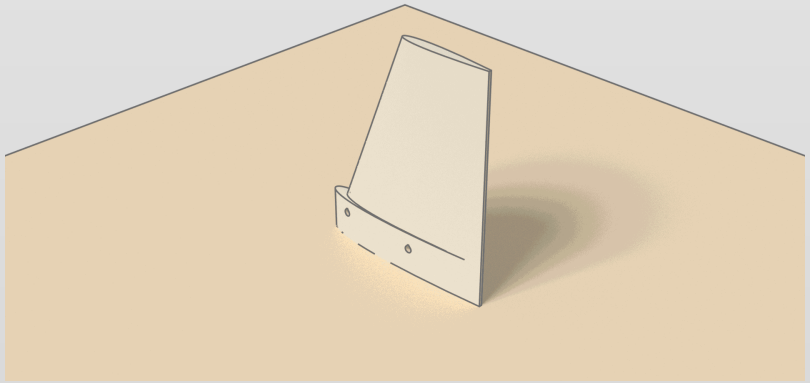
2 WALLS · 10 % GYROID · PETG-CF

M1 MOTOR PLATE

FLAT

4 WALLS · 40 % GRID · PETG-CF

PARTS ORIENTATION



FN FIN

ON ITS ROOT END

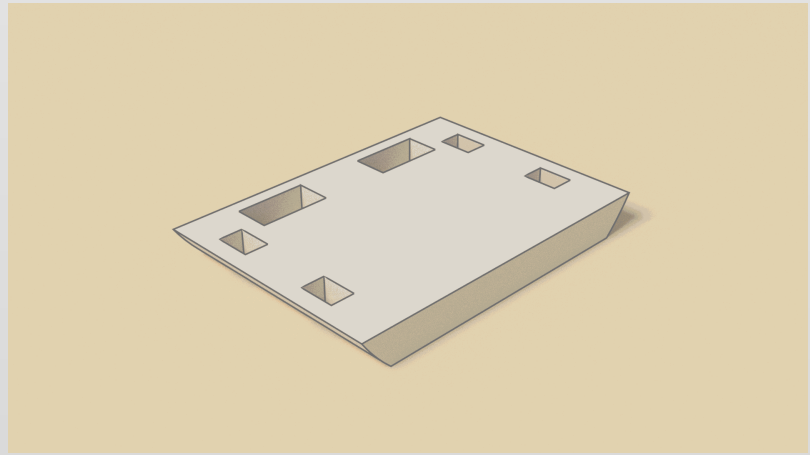
2 WALLS · 30 % LIGHTNING · ASA



SP SPINNER

BACKPLATE DOWN

4 WALLS · 40 % GRID · PETG-CF

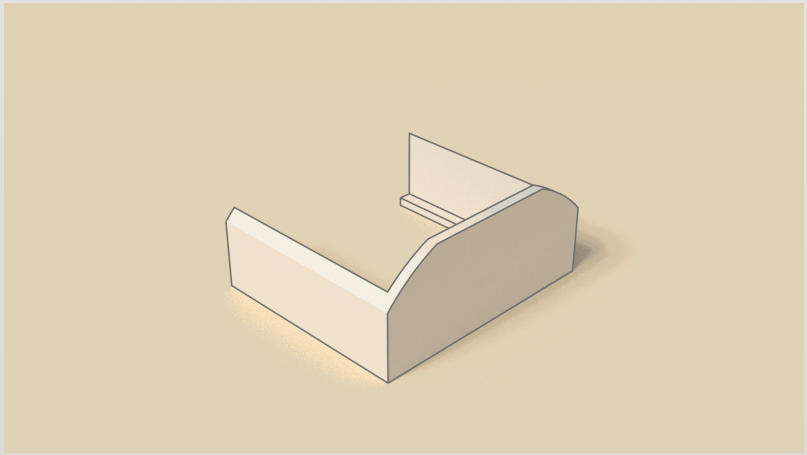


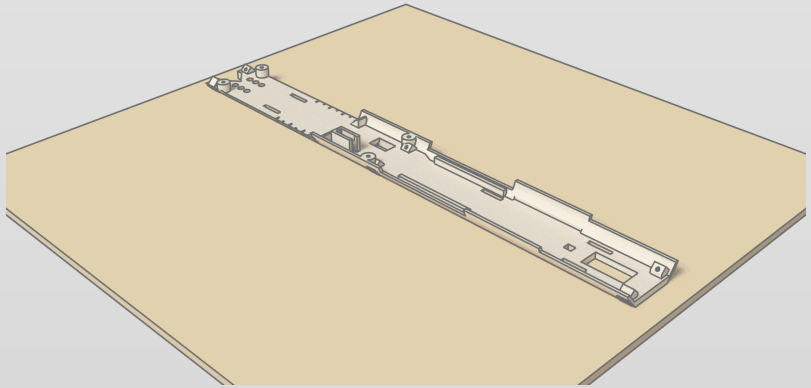
VTX SHELF (N0M)

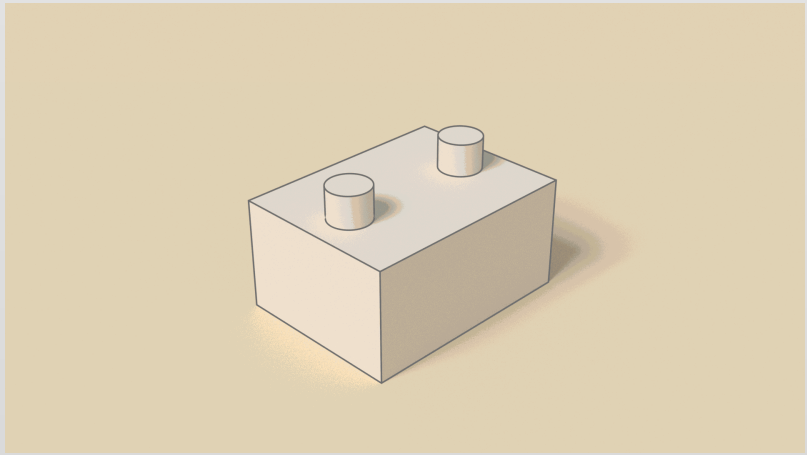
FLAT

2 WALLS · SOLID · PETG

PARTS ORIENTATION







HG GPS CARRIER

AS IT SITS, SIDE LIPS DOWN

2 WALLS · 30 % LIGHTNING · PETG

TR EQUIPMENT TRAY

UPRIGHT ON ITS FLAT UNDERSIDE

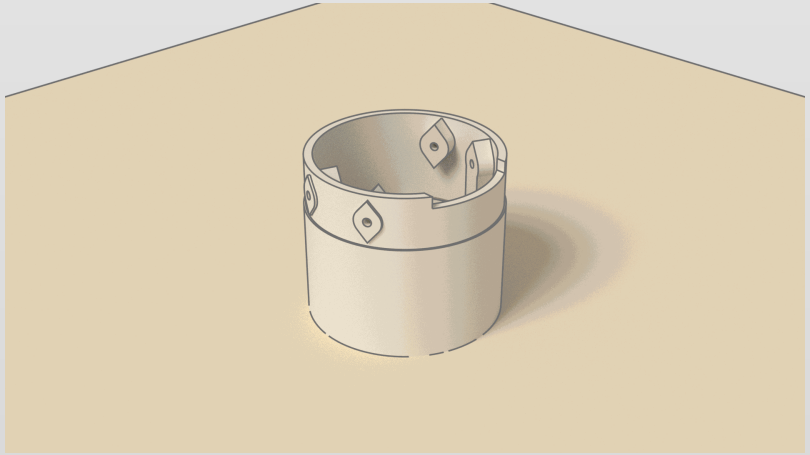
2 WALLS · 20 % GYROID · PETG-CF

BF BATTERY FRONT STOP

TOP FACE DOWN, PEGS UP

2 WALLS · 30 % LIGHTNING · PETG-CF

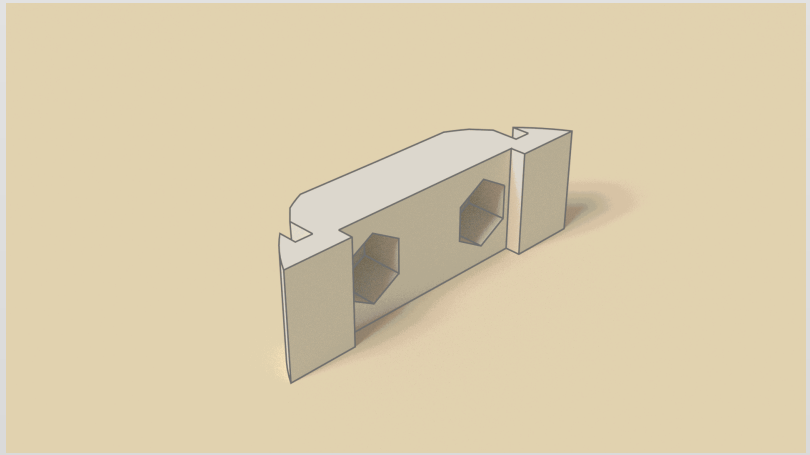
PARTS ORIENTATION



TC TAIL COUPLER

ON ITS SPIGOT END

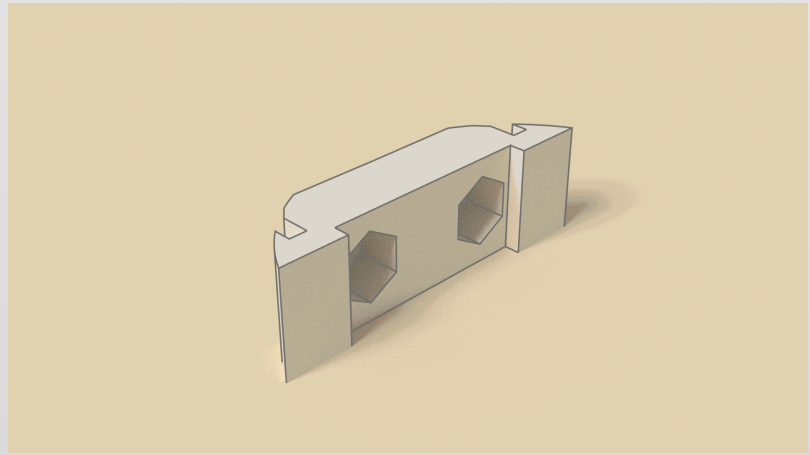
2 WALLS · 30 % LIGHTNING · PETG-CF



NB1 WING-NUT BAR FRONT

ON ITS END FACE

2 WALLS · 30 % LIGHTNING · PETG-CF

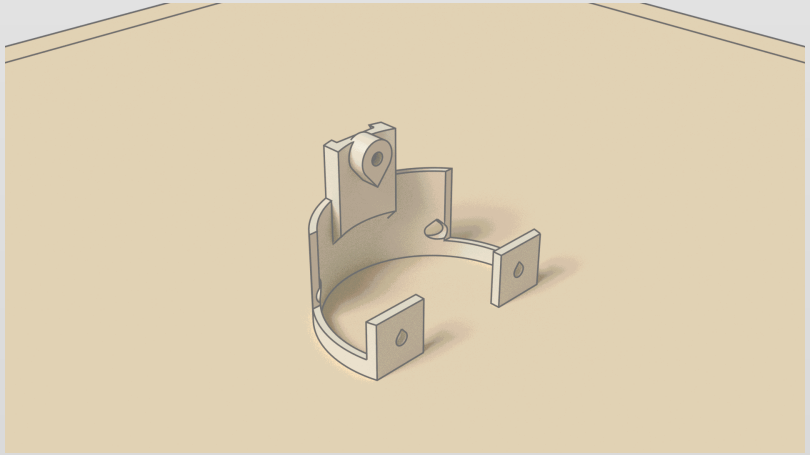


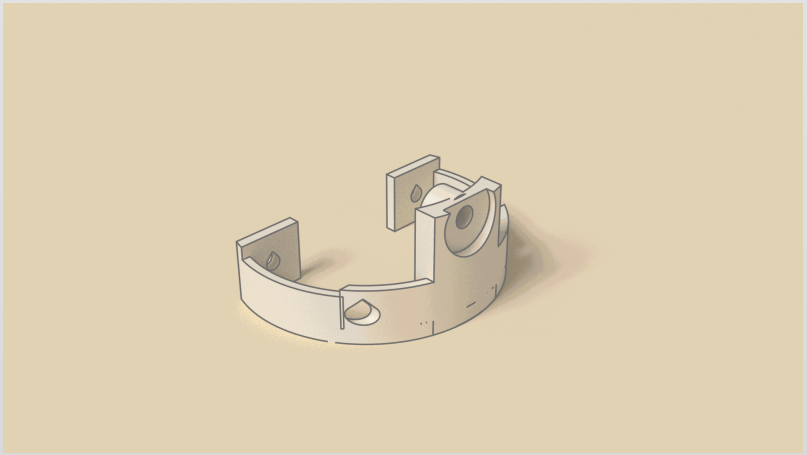
NB2 WING-NUT BAR REAR

ON ITS END FACE

2 WALLS · 30 % LIGHTNING · PETG-CF

PARTS ORIENTATION







HL1 HATCH LIP FRONT

FRONT FACE DOWN

2 WALLS · 30 % LIGHTNING · PETG-CF

HL2 HATCH LIP REAR

AFT FACE DOWN

2 WALLS · 30 % LIGHTNING · PETG-CF

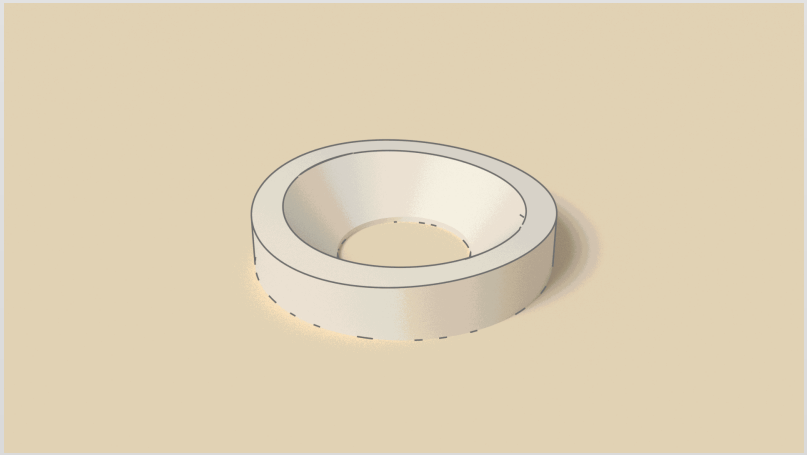
LS LIP SHIM

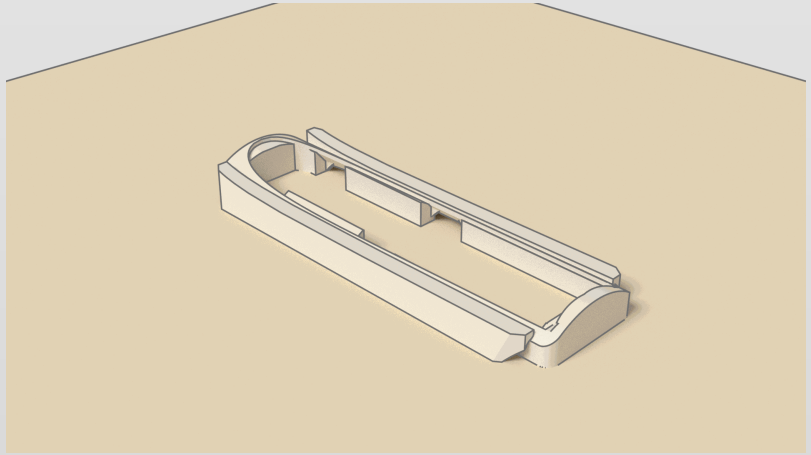
FLAT (ONE LAYER)

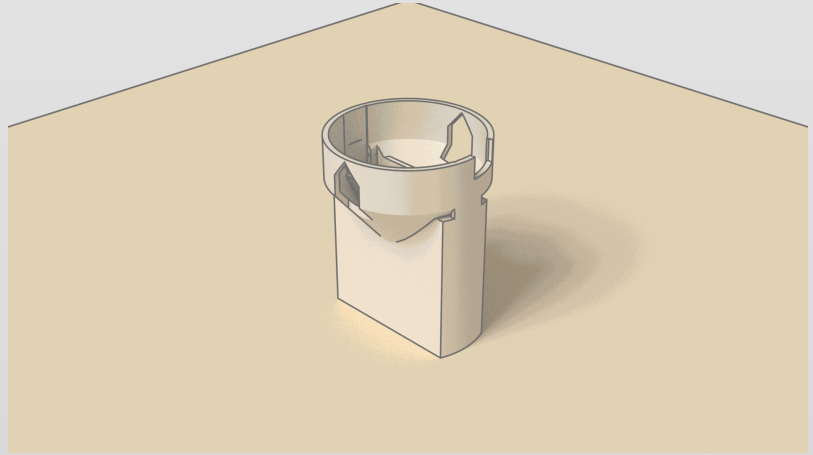
1 WALL · SOLID · PETG

PRINT 12

PARTS ORIENTATION







CW COUNTERSUNK WASHER (PRINT 18)

FLAT FACE DOWN

2 WALLS · 30 % LIGHTNING · PETG-CF

PRINT 18

WF WING-WINDOW FILLER

OUTER SKIN UP, ON ITS FEET

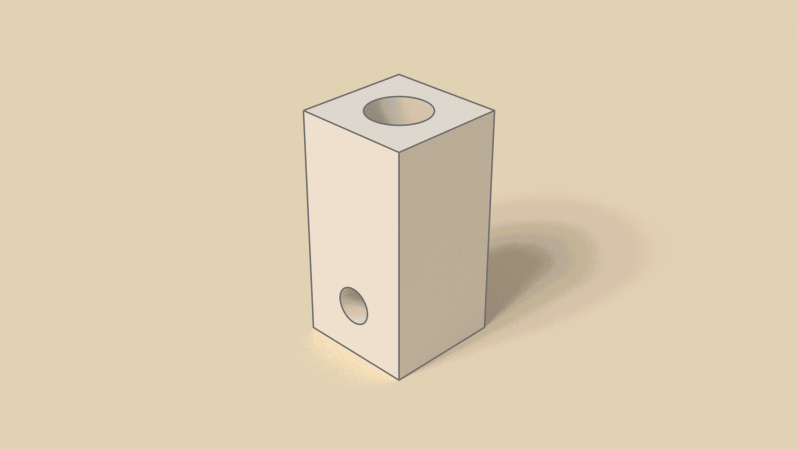
3 WALLS · 25 % GYROID · 0.12 MM LAYERS · PETG-CF

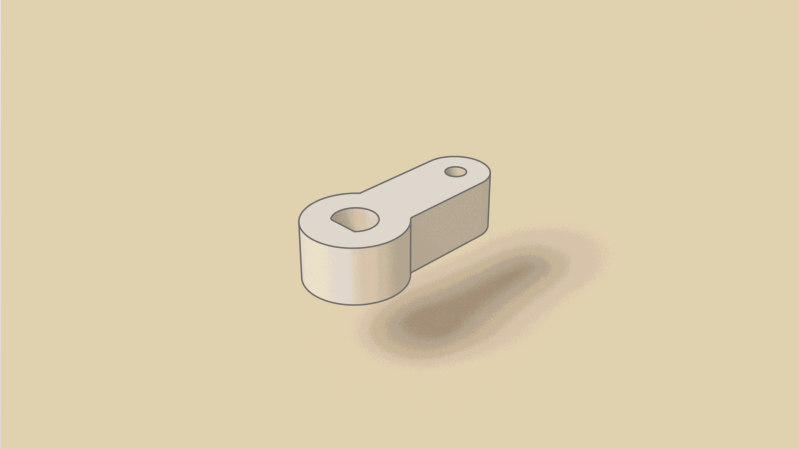
DT DJI DUCT INSERT

FRONT END DOWN

2 WALLS · 30 % LIGHTNING · ASA

PARTS ORIENTATION





AHR / AHL AILERON HORNS

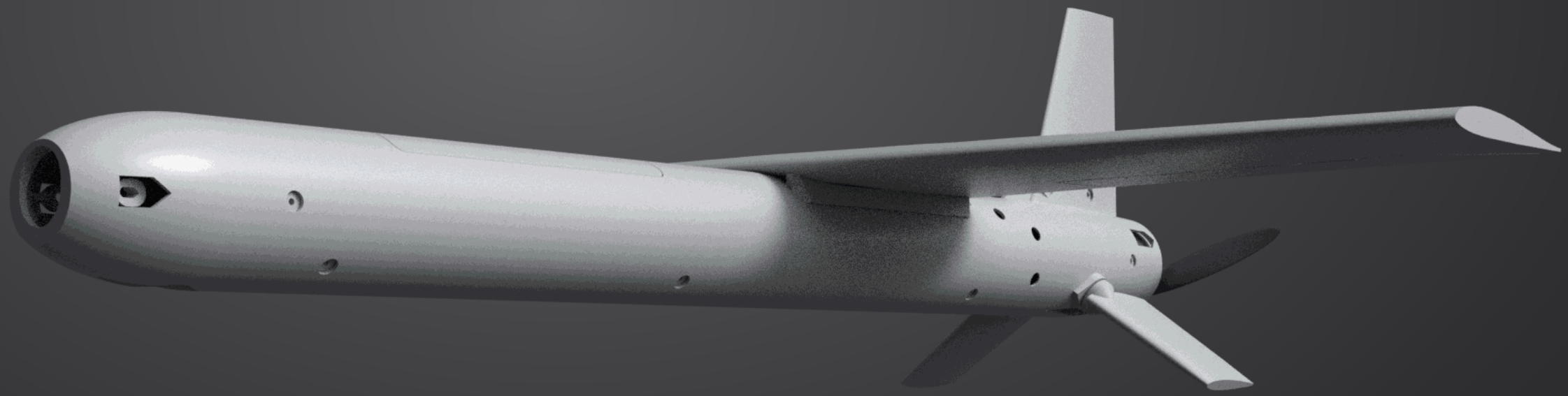
UPRIGHT, ROD BORE OPEN AT THE TOP

4 WALLS · 40 % GRID · PETG-CF

THR / THL TAIL HORNS

SHAFT BORE VERTICAL (AS THE PANELS)

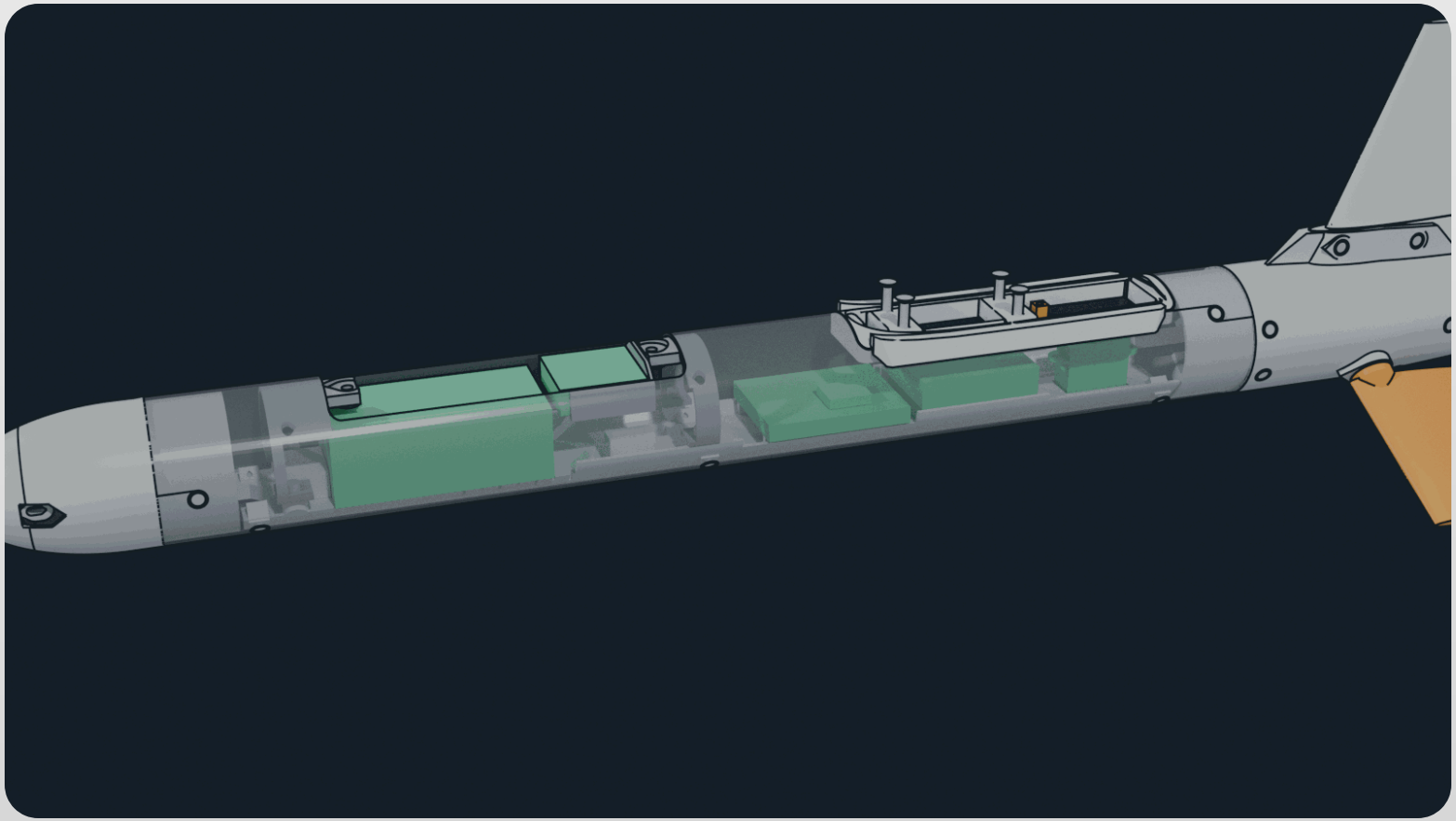
4 WALLS · 40 % GRID · PETG-CF



ASSEMBLY GUIDE

CARBON TUBE

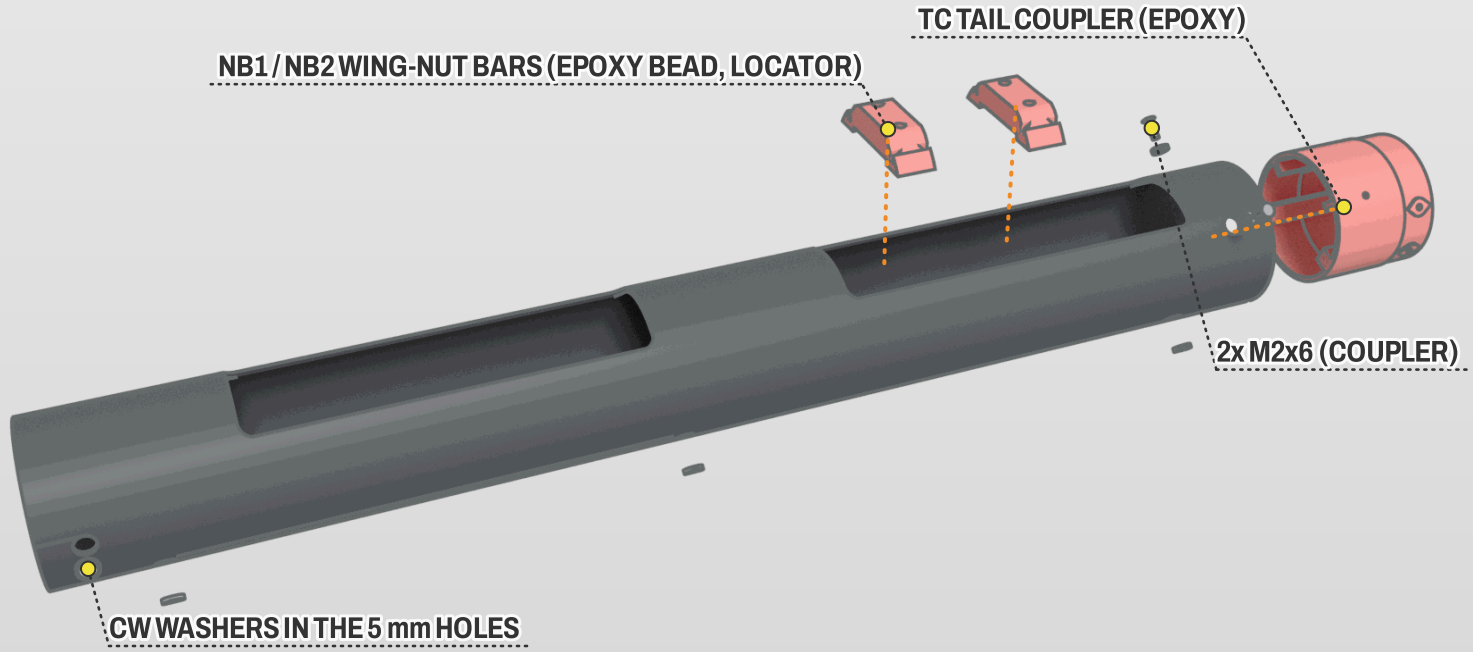
Cut the 40 × 38 mm carbon tube to 320 mm. Print the 1:1 cutting template (two A4 pages, 100 % scale: check its 100 mm bar), cut the strip out, wrap it round the tube with its ends on the tube ends and the TOP line along the top, tape it. Cut every window inside the red lines with a diamond disc and file to the line; drill the 2.2 mm holes. FFP3 mask, eye protection, vacuum at the disc or cut wet. Seal every cut edge and hole with thin CA: carbon conducts, a bare edge can cut a wire's insulation and short it.



DIAMOND DISC

CARBON TUBE

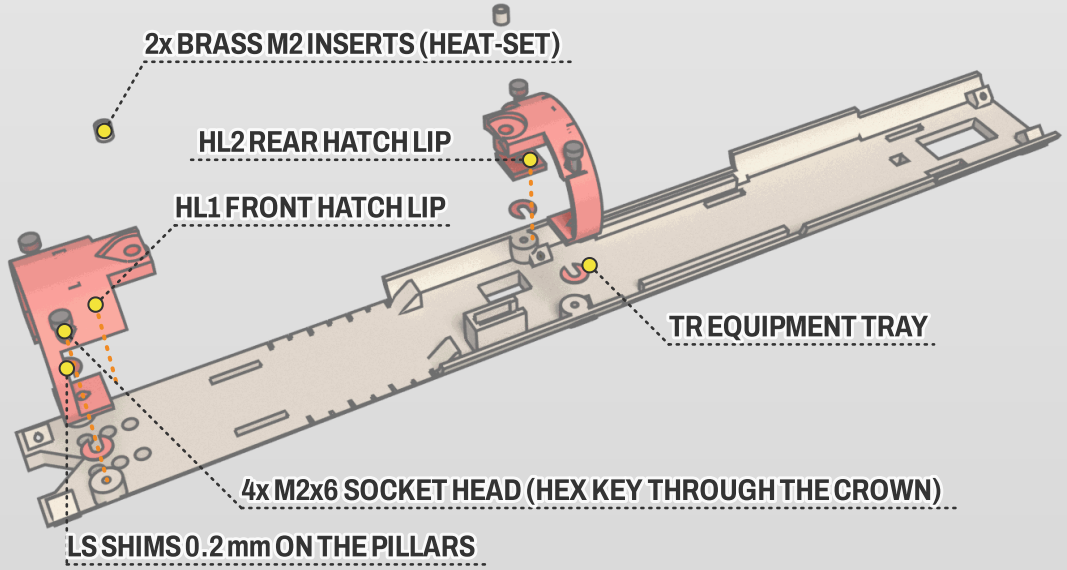
Scuff the bonding areas inside the tube (120 grit) and clean them with alcohol. With structural epoxy bond: the tail coupler TC into the aft end, its sleeve over the whole ring behind the wing window, and fix it with 2 × countersunk M2 × 6 in washers at the sides (they hold it aligned while the epoxy cures and back the bond up); the two wing-nut bars NB1/NB2 across the wing-bay window (in through the window, turned under its edges, pocket down; a thin bead only: it just holds them in place with the wing off, the wing loads go through bearing). Let it cure before the tray goes in.



EPOXY

HATCH LIPS

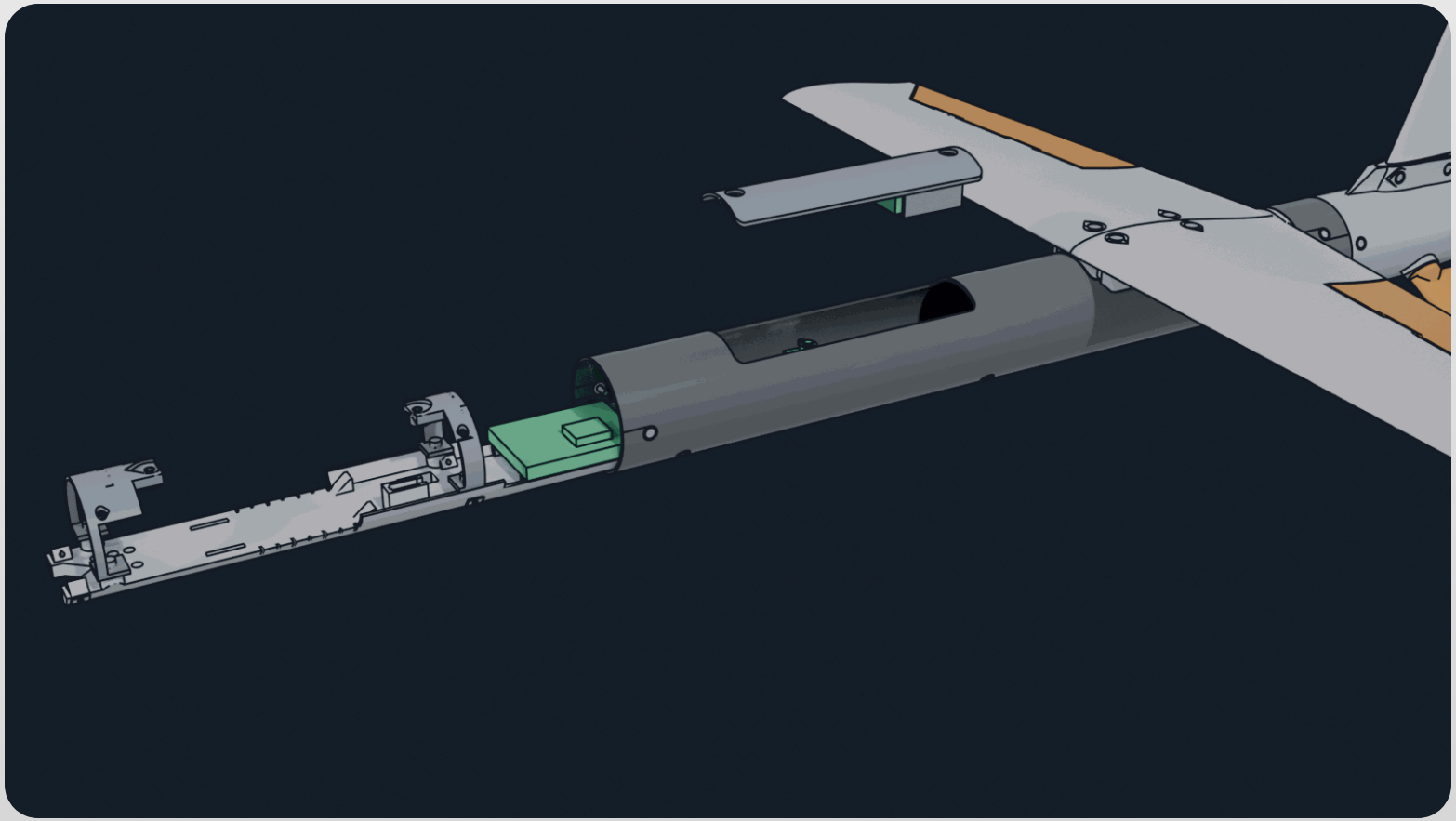
Press a brass M2 heat-set insert into each lip's cover-screw boss (soldering iron, about 230 °C), flush with the pad recess. Lay a 0.2 mm shim (LS) on each of the tray's four pillars, set the lips HL1 and HL2 on them and fix each foot tab with one M2 × 6 socket head screw (1.5 mm hex key through the hole in the lip's crown), snug only. No glue: the lips go into the tube with the tray. First fit: tray in, its screws tight, cover on; check the cover against the tube with a straight edge. Low: take the tray out and add shims under that lip's tabs (0-3 per tab); proud: remove one. The tray screws pull it onto the bore every time, so the setting repeats.



SOCKET M2 × 6 + SHIMS

EQUIPMENT TRAY

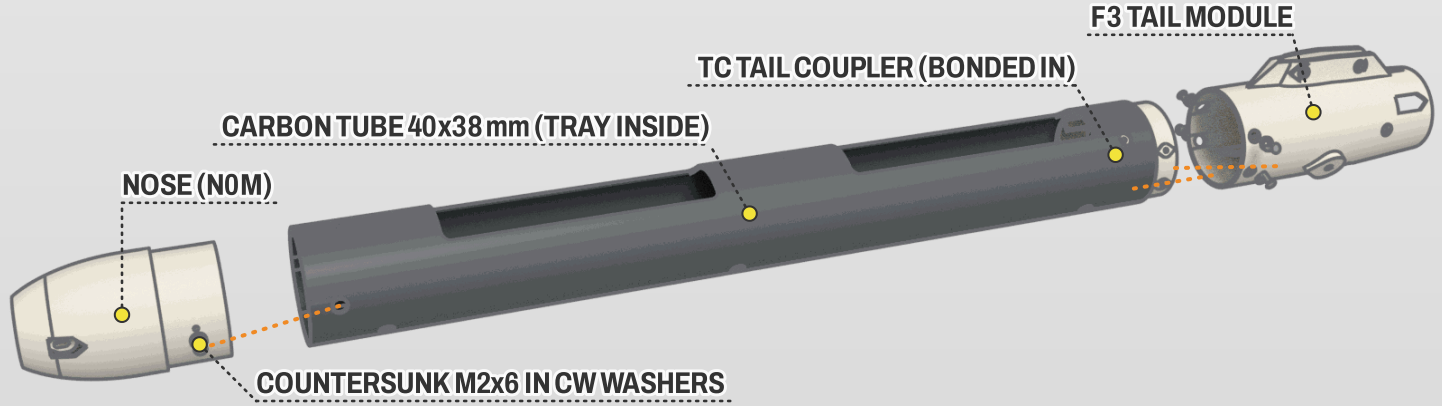
Build the tray on the bench: the ESC on its rails, the FC stack, receiver and aileron servo, all wiring as on the next pages, the hatch lips screwed on (previous page). Take the camera nose and the hatch cover off, push the tray into the tube from the front until its screw holes line up, and fix it with 6 × countersunk M2 × 6 screws through the tube (do not overtighten: they thread into printed bosses). Every external screw (the tray, the nose and tail joints, the tail coupler, the motor plate) sits flush in a printed countersunk washer (CW) pressed into its 5 mm hole: print 18, push one into each hole before the screw. To service the equipment: battery out, nose off, unhook the aileron pushrods, unplug the tail-module plugs, six tray screws out, slide the tray forward (hatch cover off: the lips come out with it).



COUNTERSUNK M2 × 6

FUSELAGE ASSEMBLY

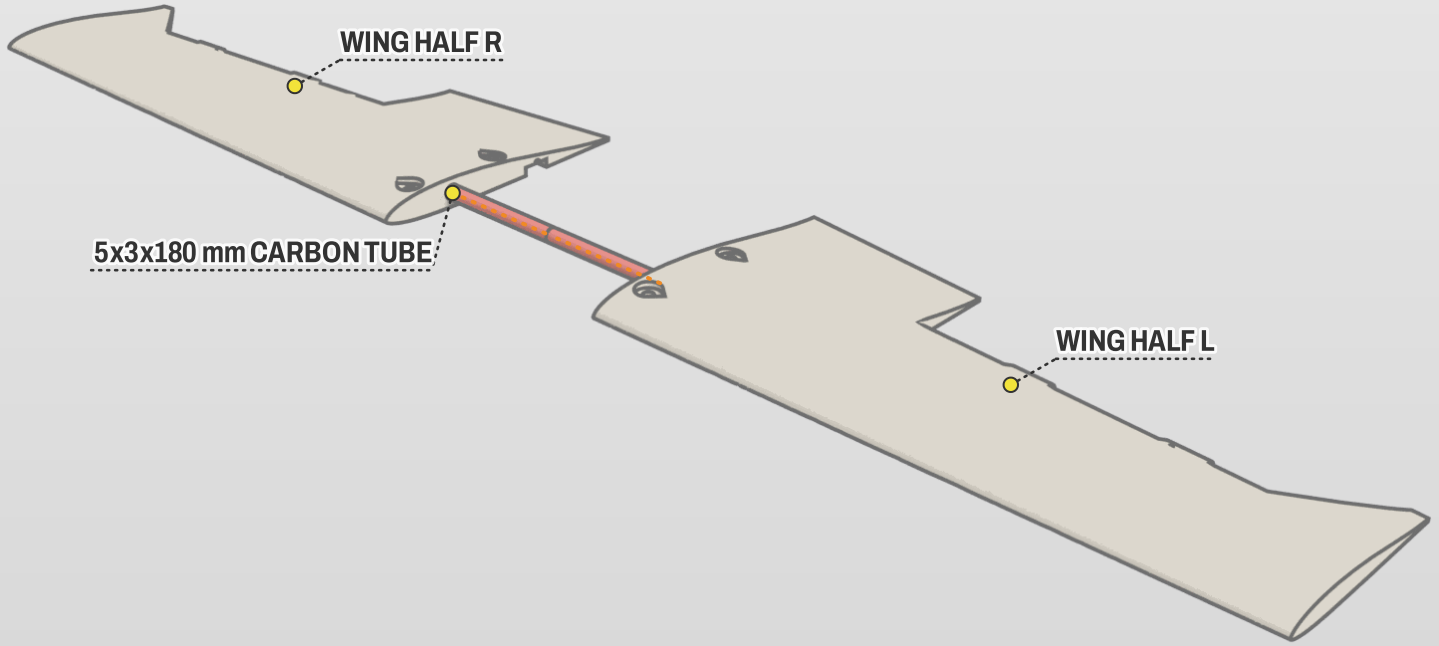
The camera nose plugs into the tube's front (its spigot is sized to the tube's bore) with 2 × countersunk M2 × 6, flush in their printed washers, into its bosses; the tail module slides onto the tail coupler's spigot (the four pads on the coupler slide in the grooves inside the module) and is held by 4 × countersunk M2 × 6 (two each side: 20° above the side line and 25° below it), each in a printed washer in its 5 mm hole, as on the tube. Nothing is glued: both modules come off for service. They must slide on by hand; if the bore is tight, sand the spigot, never force it.



M2 × 6 SCREWS

WING ASSEMBLY

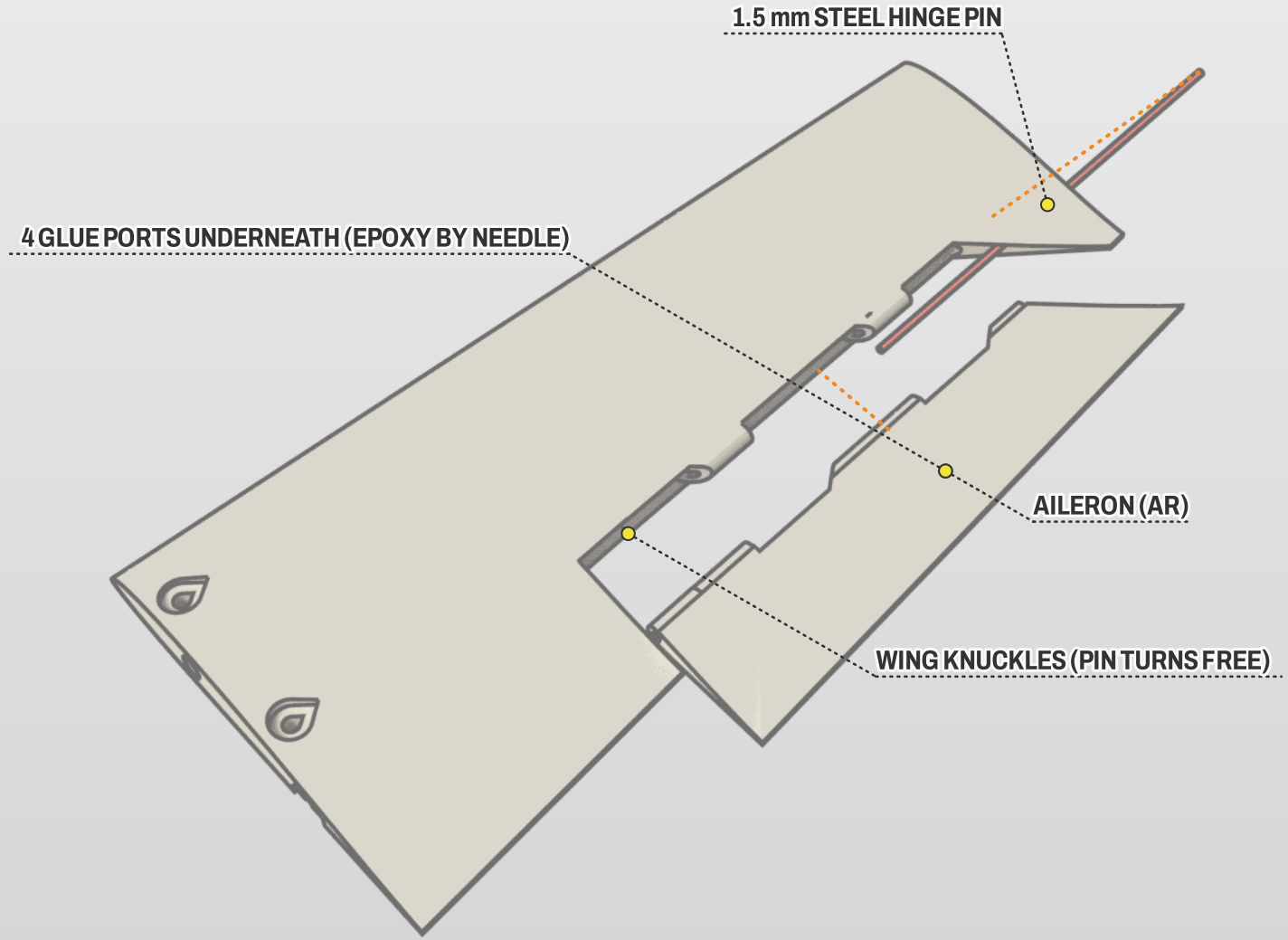
Cut one 180 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (90 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).



EPOXY

WING ASSEMBLY

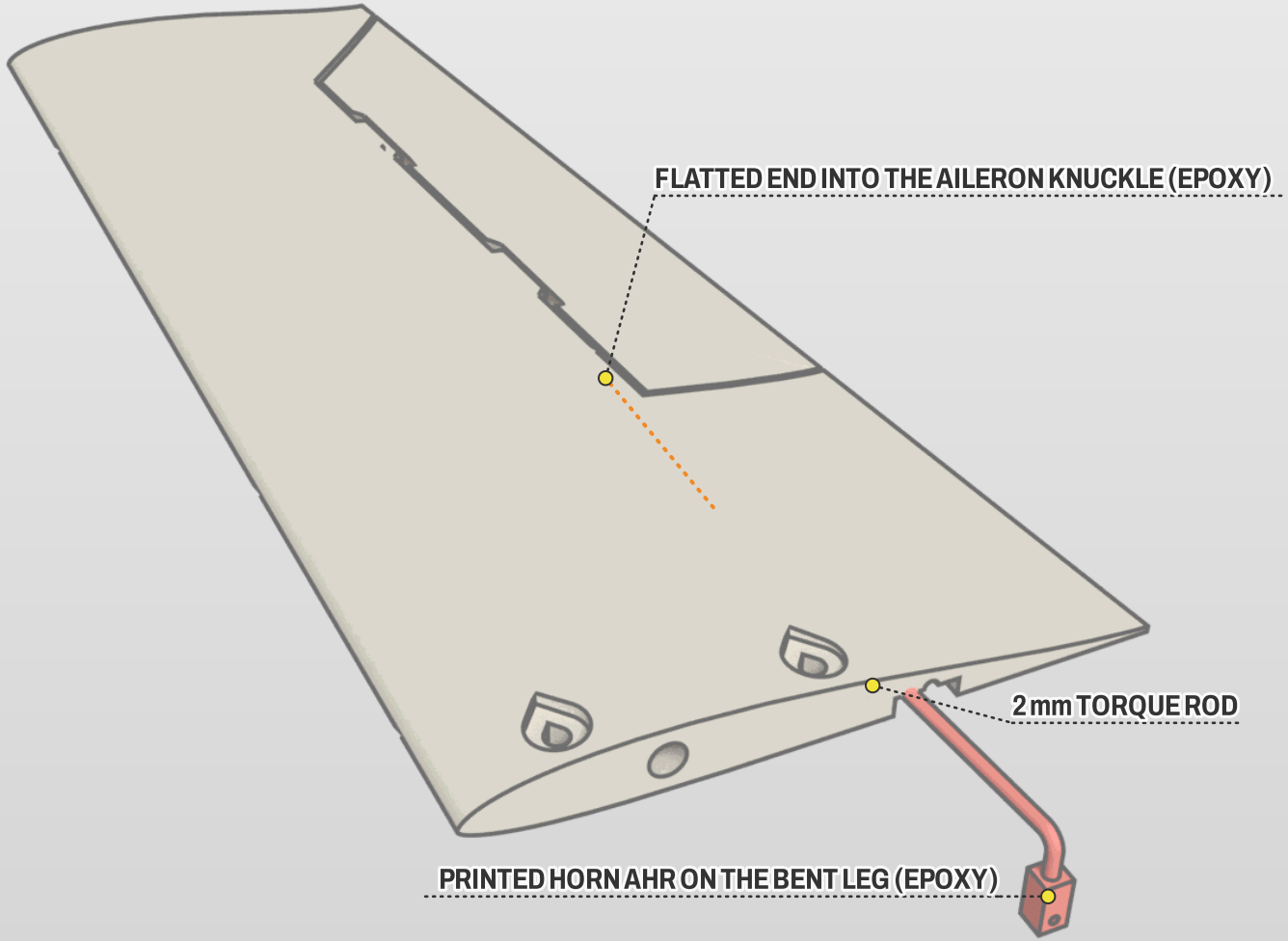
Clean the bores by hand with a drill: 1.75 mm through the wing's two knuckles, 1.65 mm in the tip. Set each aileron in place (rounded nose in the cove, knuckles in its pockets) and push a 113 mm length of 1.5 mm steel rod in from the wingtip. Dry check: the aileron swings ±20° under its own weight. Then put a drop of epoxy by needle into each of the three pin ports on its underside (none near the knuckles) and plug the tip hole.



EPOXY

WING ASSEMBLY

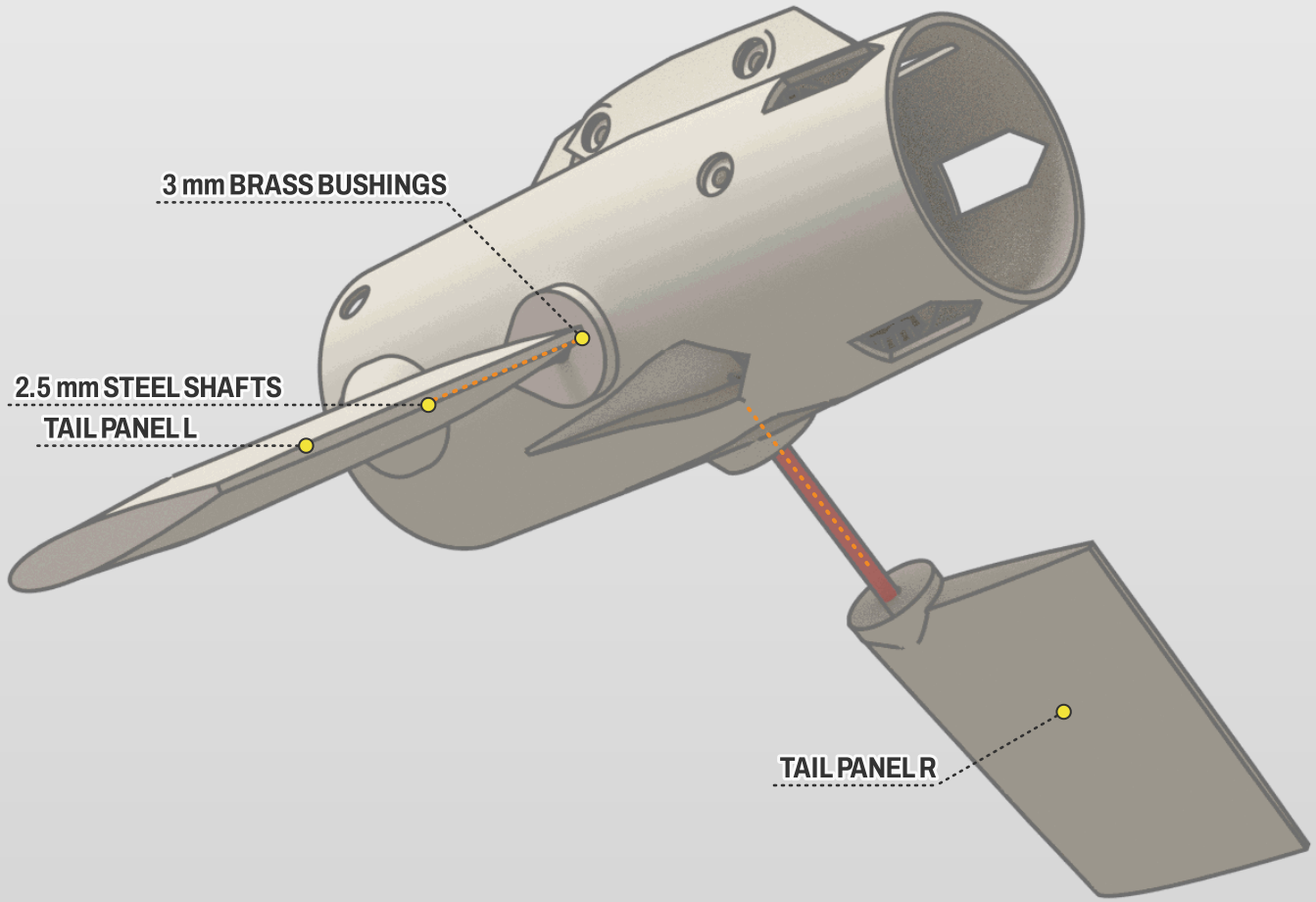
Bend each 2.0 mm rod 90° 10 mm from one end (the horn leg) and file a 0.3 mm flat on the last 9 mm of the other end, on the side away from the leg. Push it in dry from the root face, flat up and leg down through the root slot, until it seats in the D-bore of the aileron's root knuckle; then put epoxy into the knuckle's port by needle and hold the aileron 3° trailing edge up while it cures.



EPOXY

TAIL MODULE

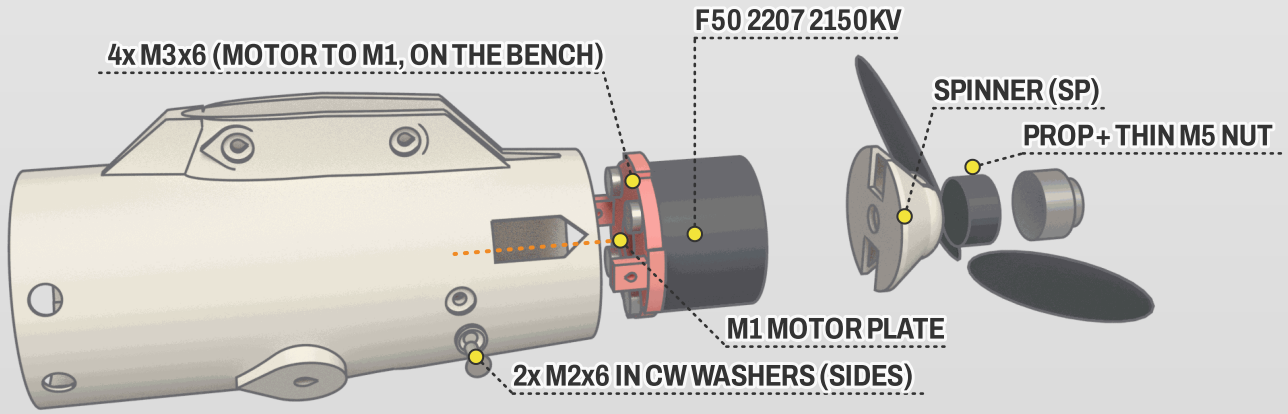
Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. Push each panel's shaft in through its bushing from outside; the printed horn (THR / THL) epoxied on its inboard end in the next step holds it, with 0.5 mm end play (no wheel collar).



EPOXY

TAIL MODULE

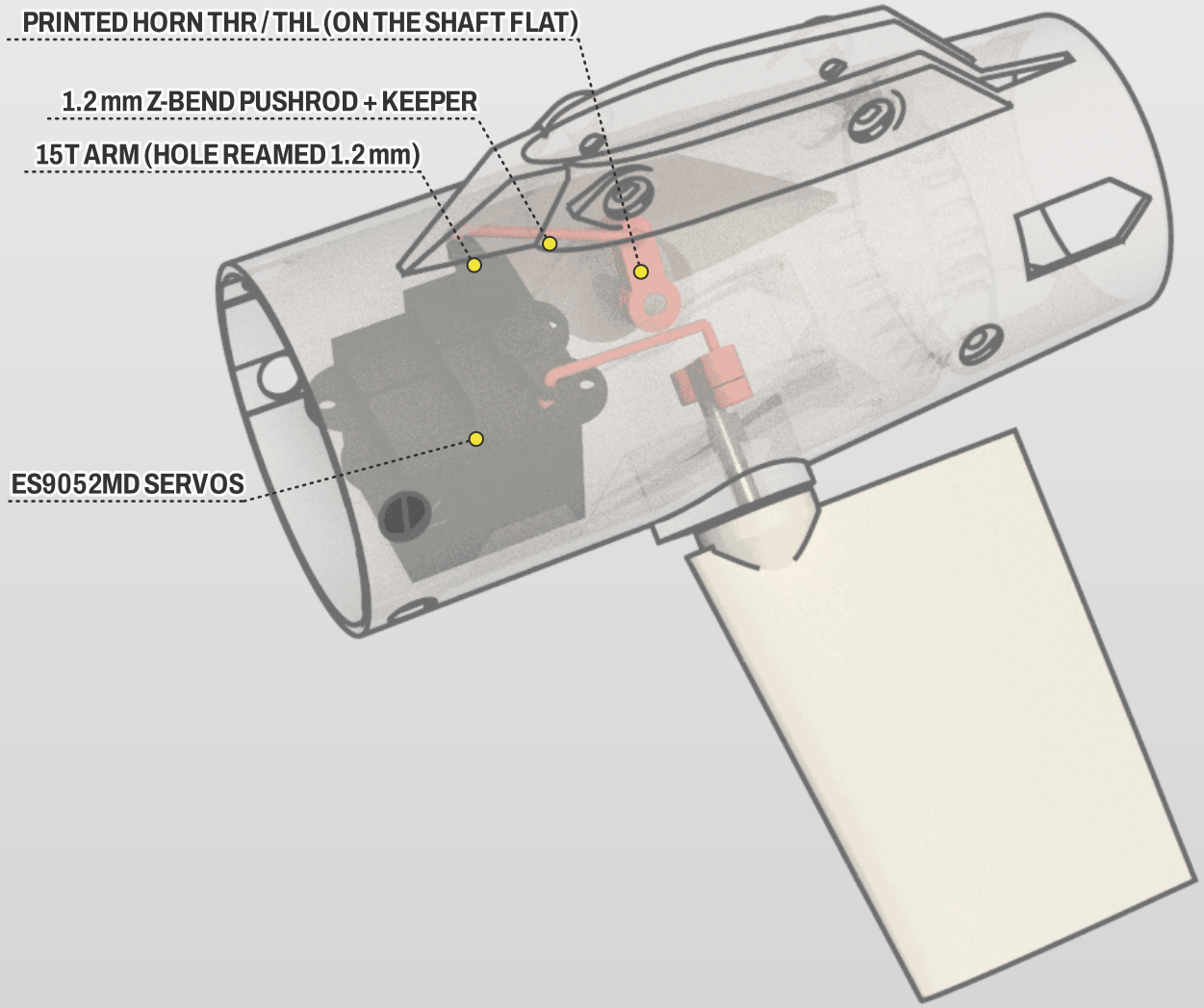
Screw the motor to the M1 plate on the bench with its 4 × M3 (heads on the plate's front face). Slide the plate into the rear of F3 until it sits on the seat ring, its two side lugs in the ring's pockets and its edge notches on the cooling windows, then fix it with 2 × countersunk M2 × 6 through the tail module's sides into the lugs, each in a printed washer in its counterbore, flush (no glue: the plate comes out for a different motor). Fit the spinner backplate, the prop (Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre polyamide, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring) and the thin M5 nut.



M2 × 6 SCREWS

TAIL MODULE

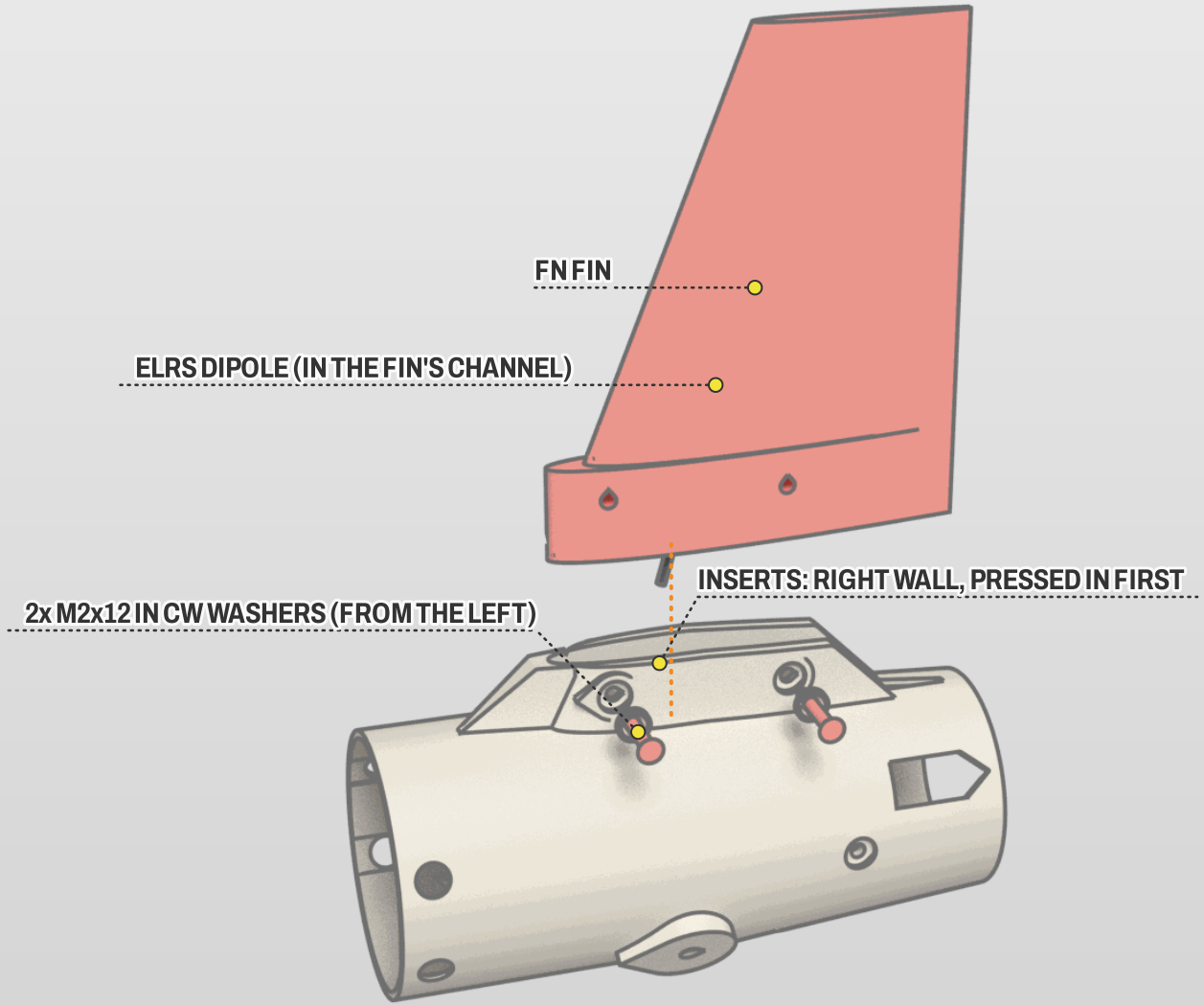
Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; file a 0.3 mm flat on the inboard end of each 2.5 mm stabilator shaft and epoxy the printed horn (THR / THL) on it, arm along the panel's normal. Drill the servo arm's 6.7 mm hole 1.1 mm and ream it by hand with a 1.2 mm bit (and the horn's hole the same): the wire must turn with no play. 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend at the horn end pushed outboard through the horn's reamed hole, snap-on keeper on the protruding end (unclip it to take the tail module off). Free play at each panel trailing edge <= 0.2 mm. Check both panels move freely through ±20°.



SERVO TAPE

TAIL MODULE

First thread the receiver antenna into the fin: the RP1 V2 T-antenna straightened into a dipole (two 31 mm legs, the feed at the centre) goes up into the closed channel inside the fin from its root, upper leg first, its coax up the parallel channel beside the lower leg; the lower leg stays 6 mm out of the root. A short strip of tape round the coax at the root face holds it up (Electrical: receiver antenna). No glue. First press the two brass M2 heat-set inserts into the blisters on the right wall of the fin shoe on F3 from outside, with a soldering iron (insert tip, ASA: about 230 °C), square and flush, and let them cool. Pass the dipole's lower leg and coax down through the slot in the shoe floor and slide the fin straight down into the shoe until its root seats. Put a printed countersunk washer (CW) in each seat on the left wall and drive the two M2 x 12 countersunk DIN 965 screws through the shoe and the fin into the inserts: snug, not tight (the walls close onto the fin). To change a fin: two screws out, lift it off.

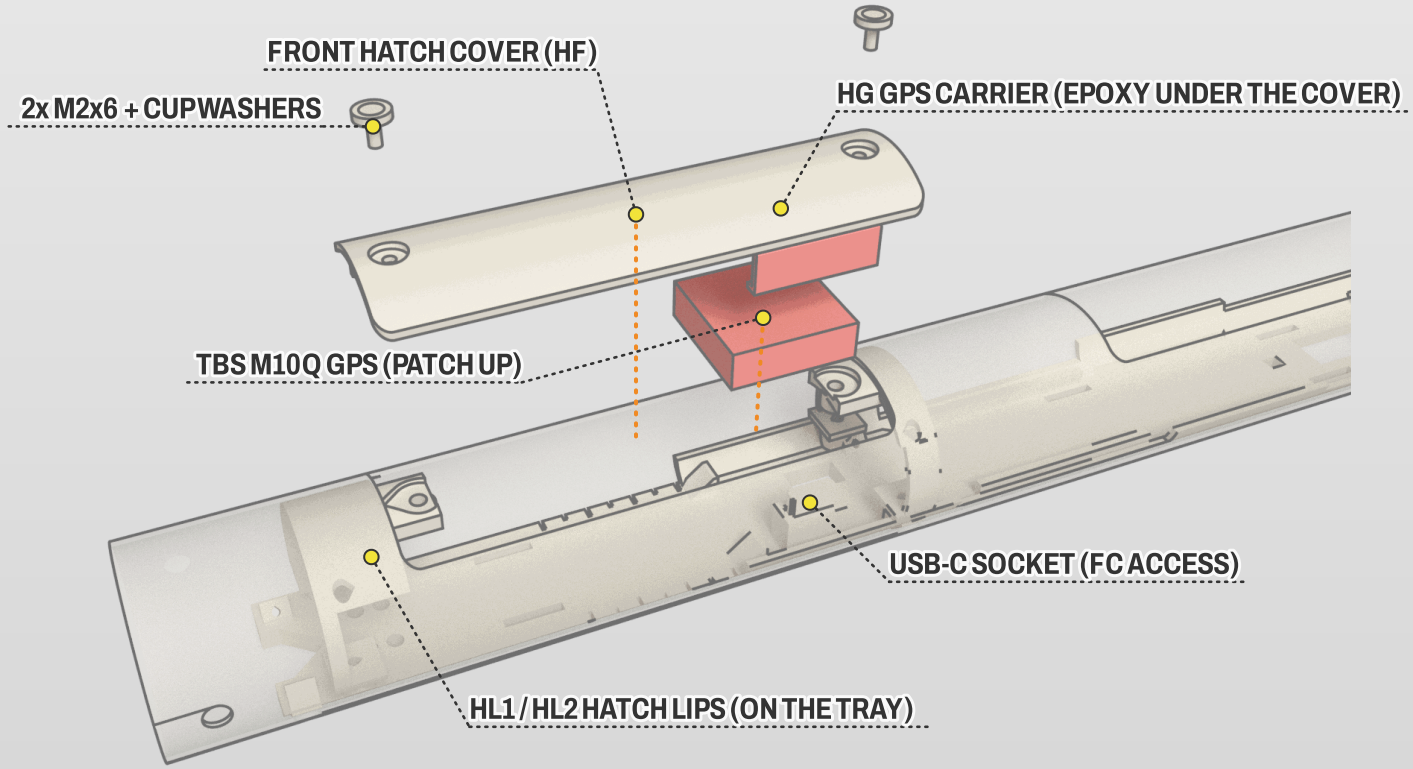


2 × M2 × 12 + INSERTS

FUSELAGE ASSEMBLY

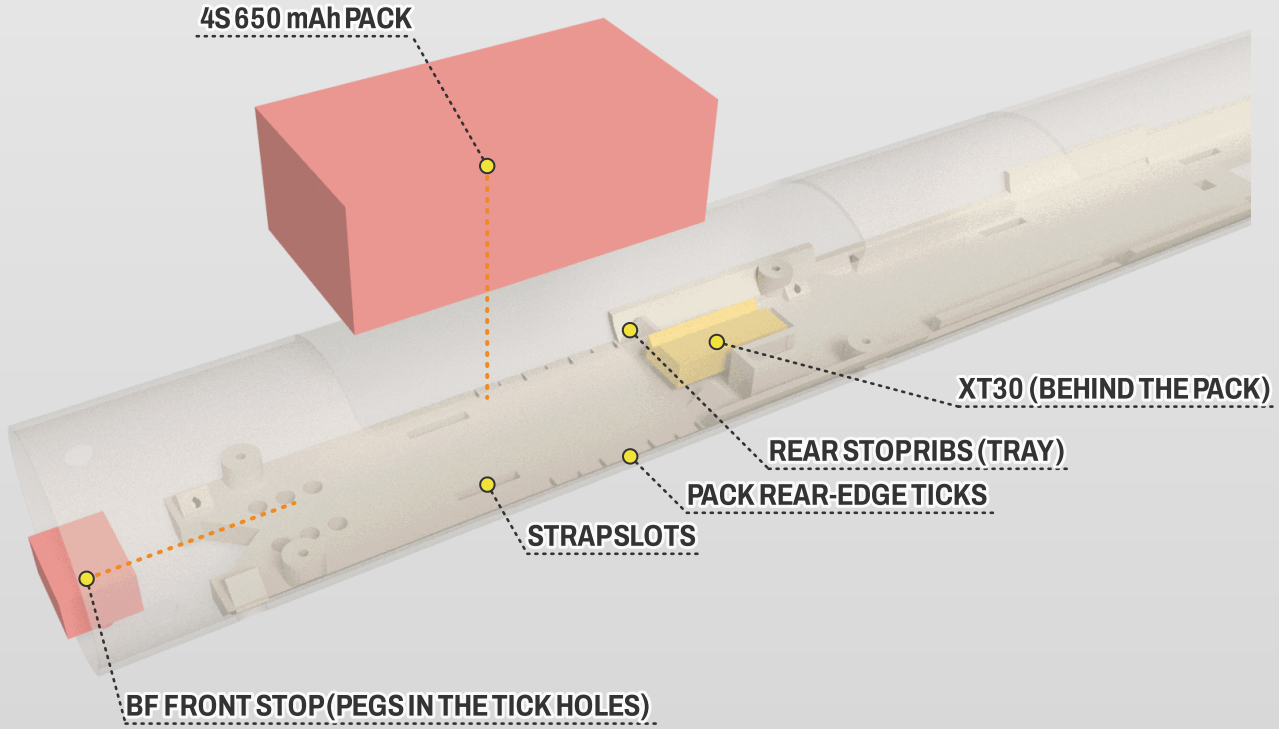
Bond the printed GPS carrier (HG) under the rear of the PETG hatch cover (the GPS's radio window: never paint it with metallic paint). Put foam tape on the carrier's two side lips and slide the TBS M10Q GPS in from the front, patch up, onto the lips against the rear wall: it is carried from below, with nothing but air and the cover between its patch and the sky. The cover is held by two flush countersunk M2 × 6 screws in steel cup washers, one at each end, into the brass inserts in the lips' bosses (the lips are screwed to the tray; no magnets: suction and the GPS's weight would lift them at speed). Leave a slack loop in the GPS lead so the cover lifts off. With the cover off you reach the battery, its XT30 and the FC's USB-C socket (left side, beside the GPS lead): plug the laptop in there for setup, parameters and logs. The socket board presses into its holder on the tray before the tray goes in; its extension lead leaves through the slot in the holder's aft wall and runs along the floor to the FC's USB-C port.

FOAM TAPE



FUSELAGE ASSEMBLY

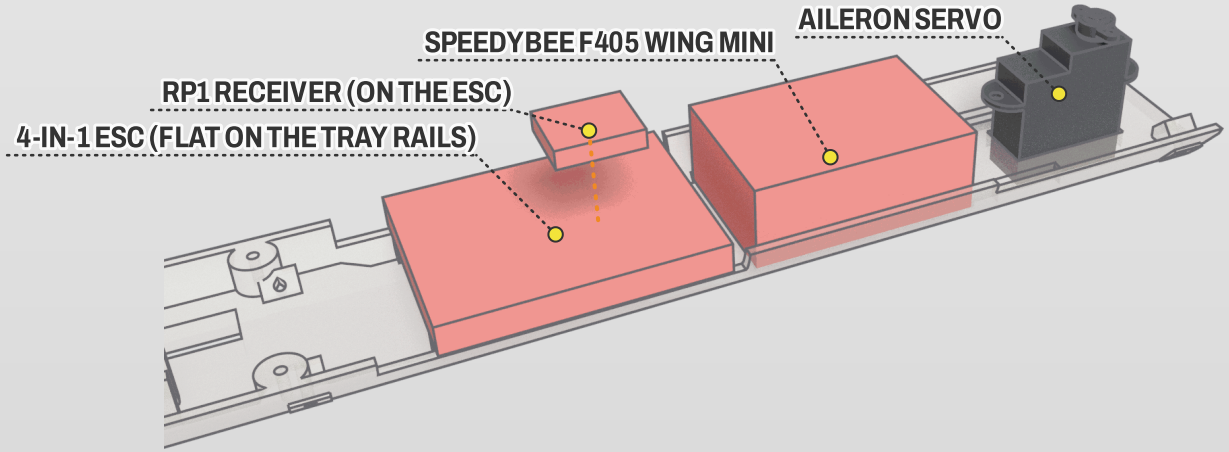
Lower the pack through the hatch against its rear stop (two ribs on the floor at x 160: the pack's rear edge), then slide it forward under the tube until its rear edge is at the tick the nose table gives (ticks every 5 mm on the floor edges; pack centre = tick − 30 mm). With the N0M nose the balance position is the rear stop itself: rear edge at tick 158, pack centre x 128; the printed front stop (BF) then holds it from the front: fitted once on the bench, nose off, its two pegs in the floor holes for that tick (holes for ticks 150, 155 and 160). One strap through the slots. The XT30 pair lies behind the pack under the GPS and unplugs through the hatch. To take the pack out, slide it back against the rear stop (under the hatch opening) and lift it out.



VELCRO STRAP

EQUIPMENT

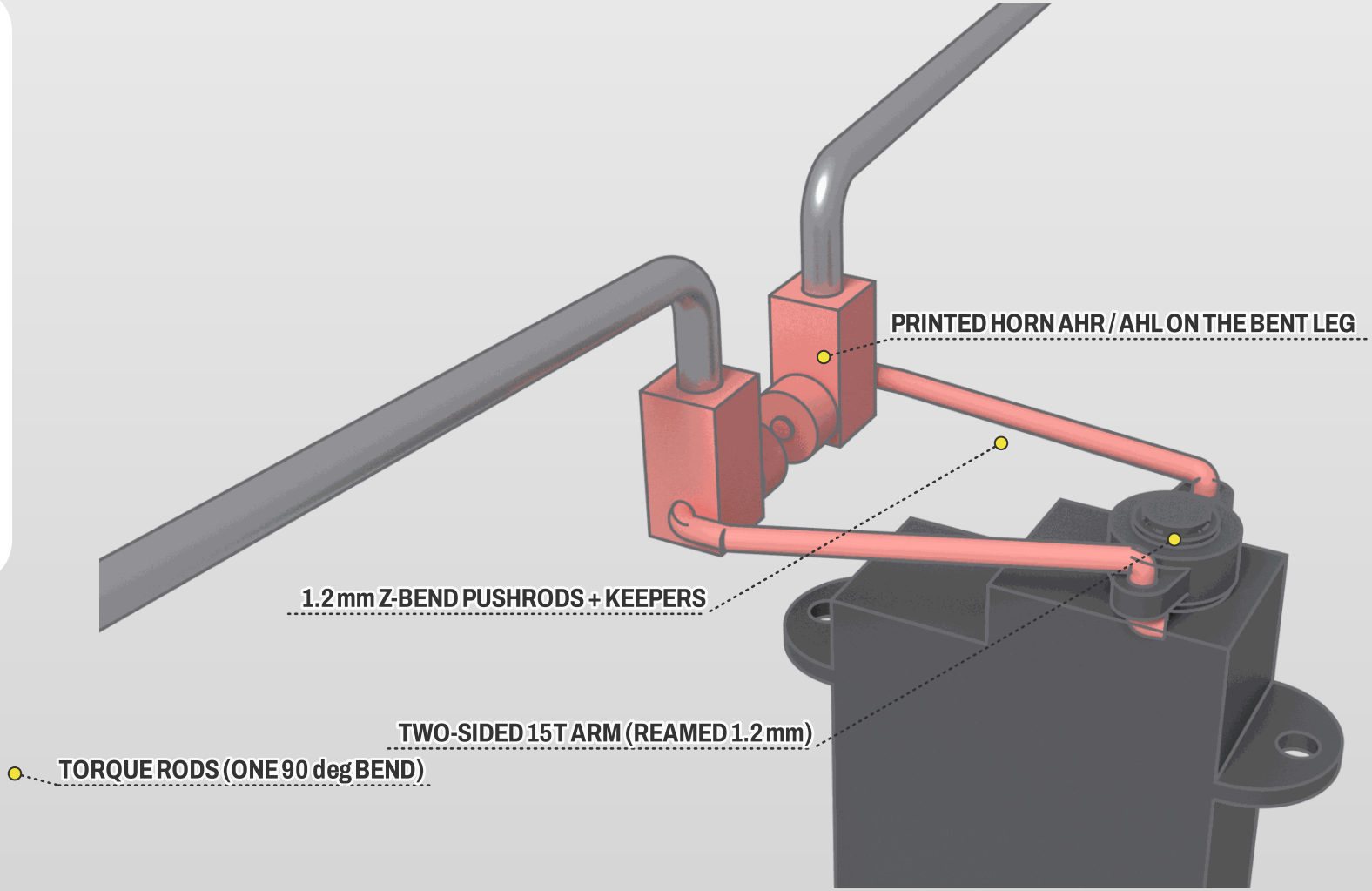
Wing bay, front to back: the ESC flat at the bottom (x 217–260) on foam tape on its two rails, component side up; the receiver on top of the ESC on foam tape; the SpeedyBee F405 WING MINI stack 5 mm behind the ESC (x 265–302) on a 0.5 mm foam pad and a strap, arrow forward, servo header row on the right, servo BEC jumper at 6 V. Aileron servo aft of the stack, output at the aft end. The receiver's antenna is the dipole in the fin (Electrical: receiver antenna).



FOAM TAPE

EQUIPMENT

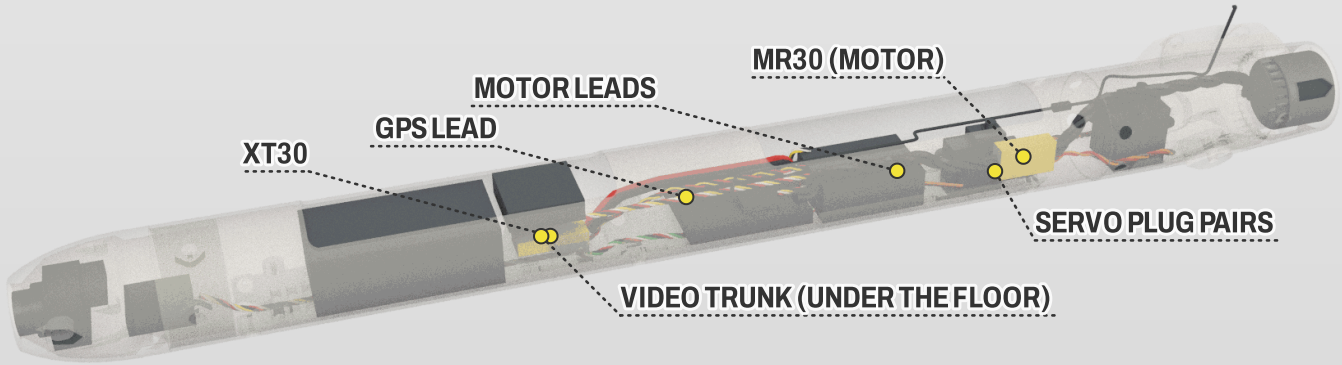
ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque rod has one 90° bend (the horn leg, 6 mm from the centreline); epoxy the printed horn (AHR / AHL) on the leg, 4 mm deep. Drill the arm's 4.3 mm holes 1.1 mm and ream them by hand with a 1.2 mm bit (and both horns' holes): the wire must turn with no play. Each horn connects to the arm with a 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend pushed inboard through the horn's reamed hole, snap-on keeper on the protruding end. Check: with the servo powered, each aileron trailing edge moves <= 0.2 mm (free play lowers aileron flutter to about 33 m/s). Set both ailerons 3° trailing edge up at servo neutral. No play at either end.



Z-BENDS

EQUIPMENT

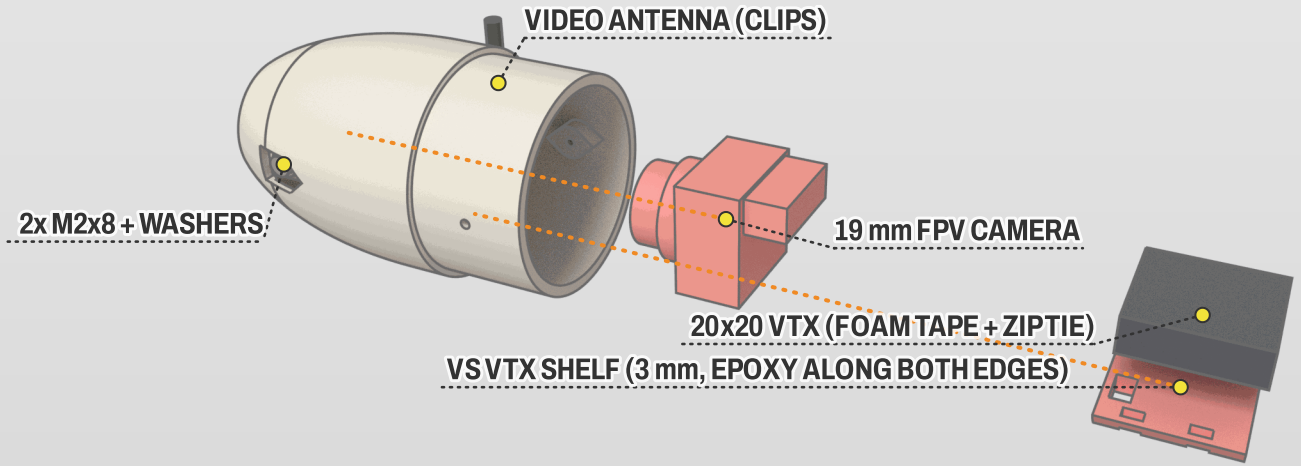
ESC flat on its foam tape at the bottom of the wing bay, receiver on top of it, FC stack 5 mm behind it (servo header row on the right). Battery lead (14 AWG) from the XT30 over the ESC and down the gap to the FC power board's BAT pads; 14 AWG ESC feed from its ESC pads up onto the ESC; ESC signal (M1, GND, TLM) and receiver lead up the gap to the FC; motor phases up the gap, over the FC's left half to the MR30; the ELRS coax under the wing-nut bars to the u.FL joint ahead of the tail joint and up into the fin. Cut lengths and lanes: the Electrical integration chapter of this manual. Unplug the MR30, the u.FL joint and the servo plugs before pulling the tail module.



HEAT SHRINK

CAMERA NOSE

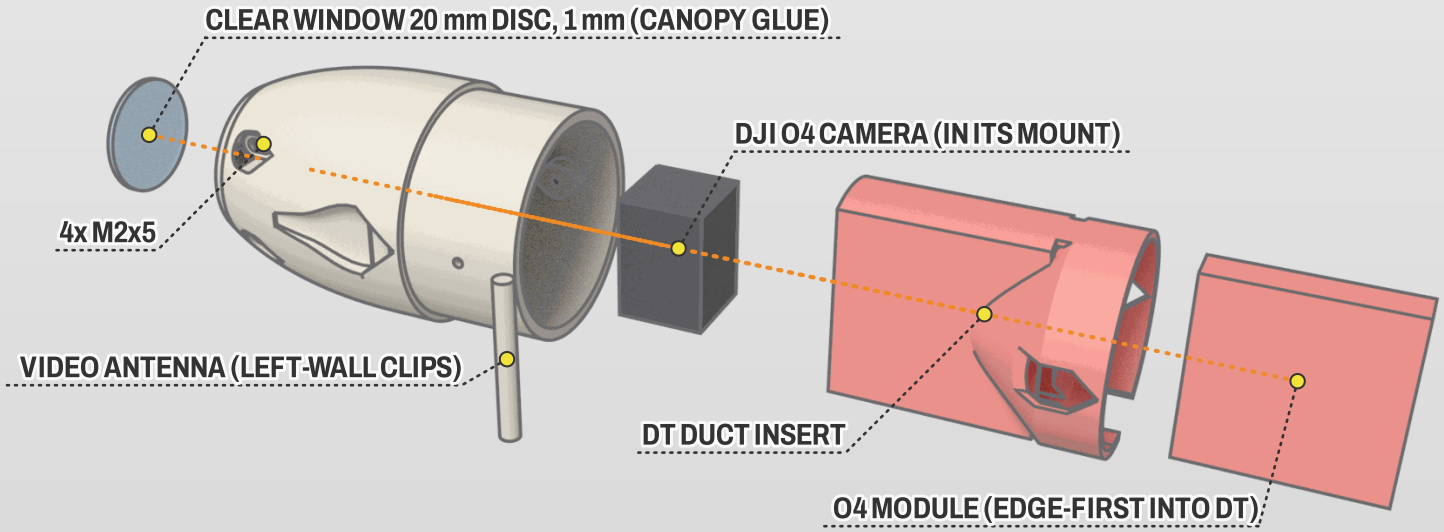
Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the 3 mm shelf in from the rear until its curved side edges seat in the spigot bore (VTX on top), then glue both edges with a thin bead of epoxy; video antenna in the two clips on the right wall. Plug the one lead at the nose joint.



FOAM TAPE

CAMERA NOSE

DJI O4 option, in this order: (0) cut the clear window disc (20 × 1.0 mm) with a circle cutter, drop it into the face recess from the front (the recess centres it; it sits flush), lift it out, put a thin bead of canopy glue or clear RTV on the seat ring only (no CA: it fogs the window), press the disc back in and tape it flush while it cures (the notch at the bottom lets a knife tip lift it out later); (1) snap the video antenna into its two clips on the left wall, from the rear; (2) slide the camera in its mount in from the back and clamp it with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount), lens flush; (3) plug the coax and the 3-in-1 lead into the module and slide the module edge-first into the duct insert (DT) until it stops, foam strip in the top groove; (4) slide the insert in through the spigot until its two notches stop on the joint-screw bosses. The tray holds it once the nose is on the tube. Cooling air comes in through the two side NACAs and rises over the insert's sill before it reaches the module: never tape the module's faces.

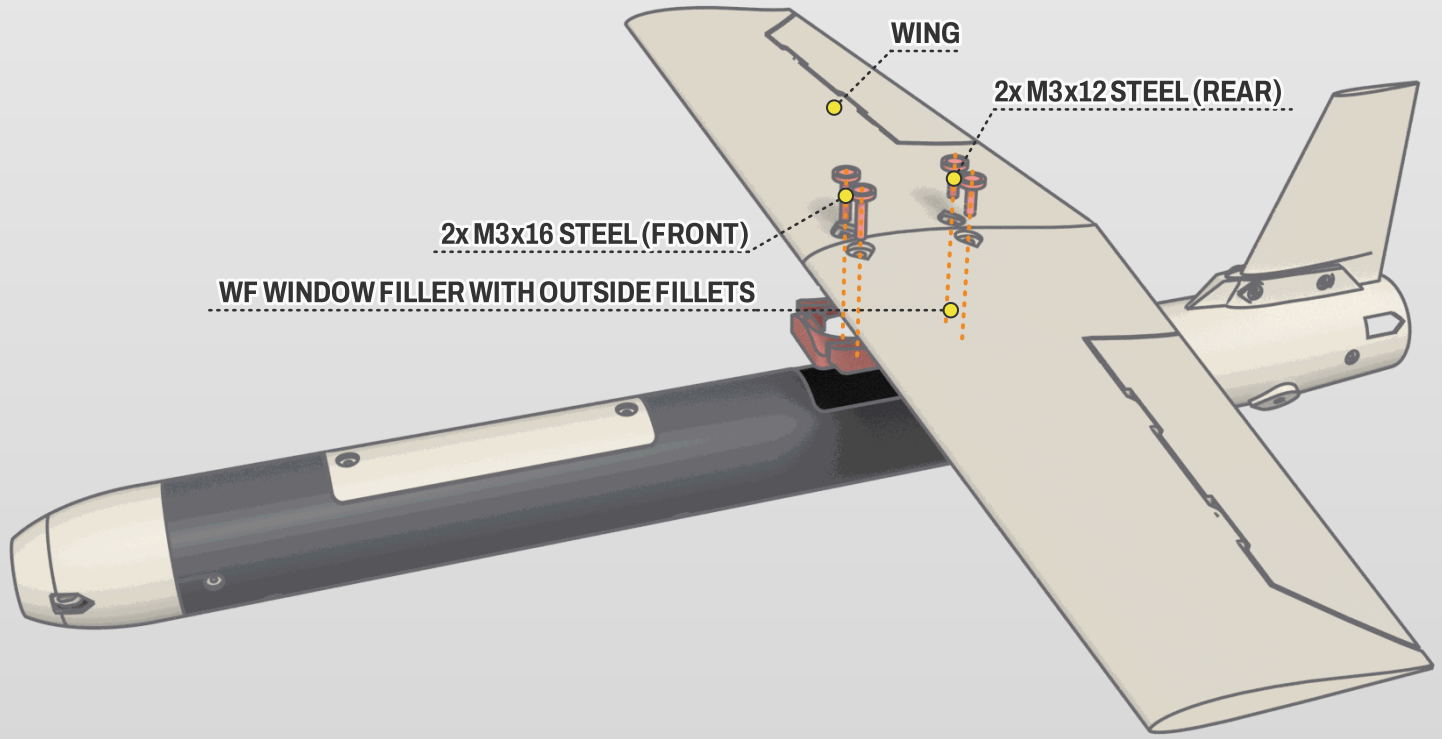


M2 × 5 SCREWS

FINAL ASSEMBLY

Push an M3 nut up into each of the nut bars' pockets (two per bar). Set the printed window filler (WF) into the tube's window, no glue: its skin strips drop into the window and its two outside fillets rest on the carbon along both window edges. Set the wing on it and press it down: it must sit on the nut bars and on both fillets without rocking. Check with paper strips under each fillet at the front and the rear; sand the fillet tops (or add 0.1 mm tape) until the wing sits firm with no play. Bolt the wing with the steel countersunk screws in steel cup washers (flush in the recesses), two at each station, one in each wing half (each half is clamped on its own, off the glued seam): M3 × 16 at the front, M3 × 12 at the rear; they pull the nut bars up against the tube from inside. Snug them with a small driver, do not overtighten (the nut bars are printed). The screws hold the filler between the wing root and the tube; the bars bear on the inside of the carbon and the fillets on the outside (no glue carries the wing). The filler comes out with the wing off, for sanding or a new one.

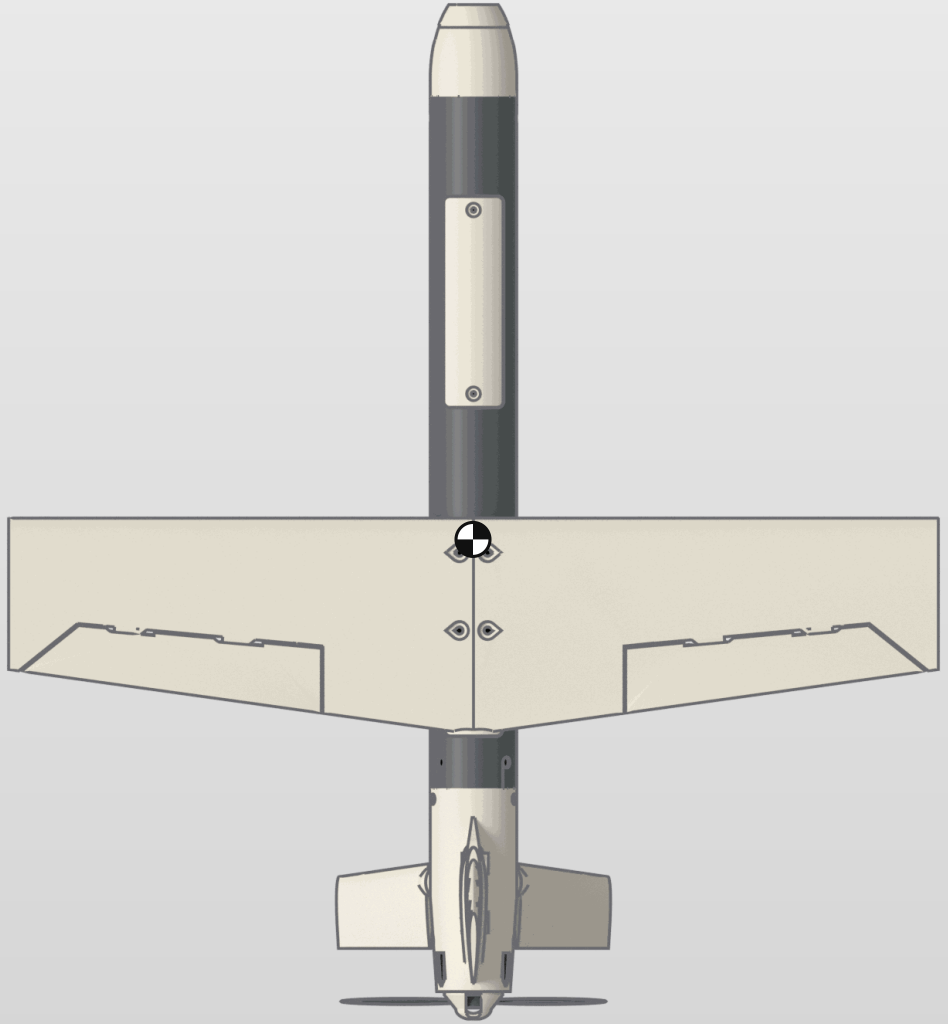
STEEL SCREWS



FINISHING BUILD

Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws ±10° on the tail panels, about ±10° (±3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

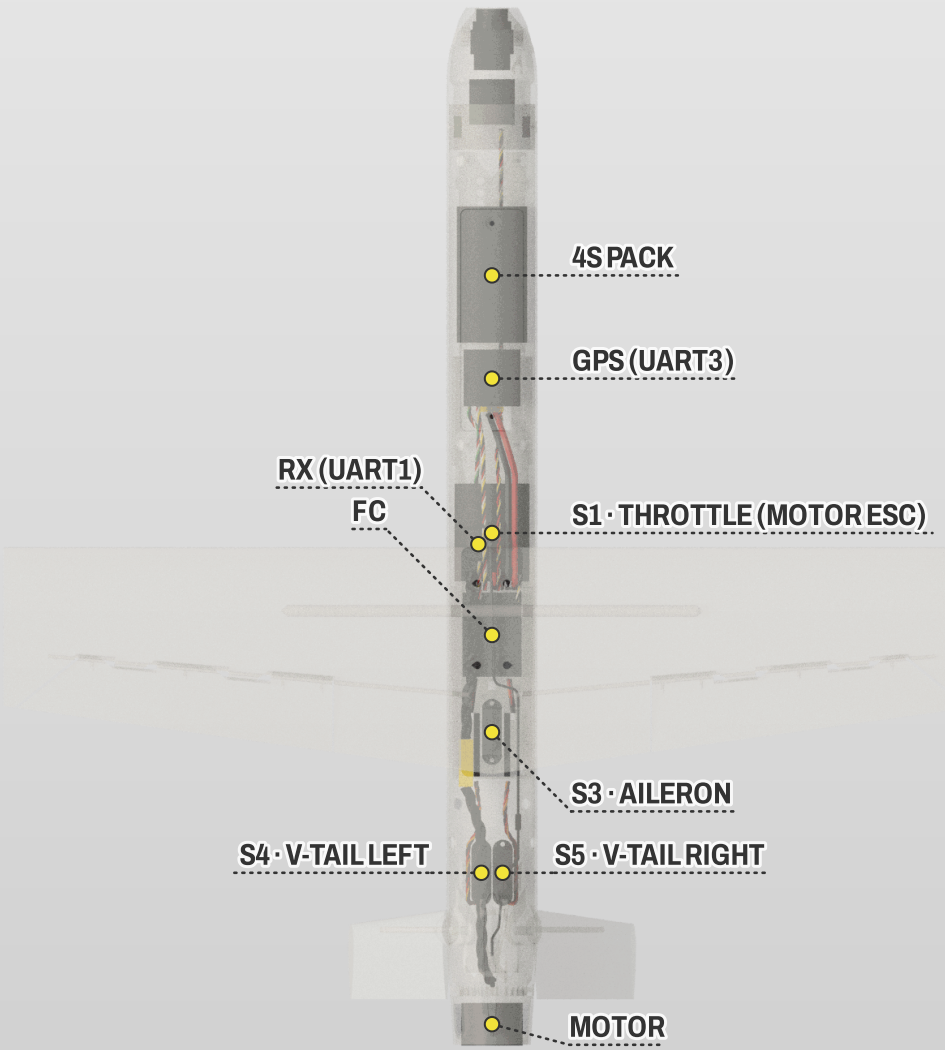
Balance: CG 10 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose ballast cup (N0P nose).



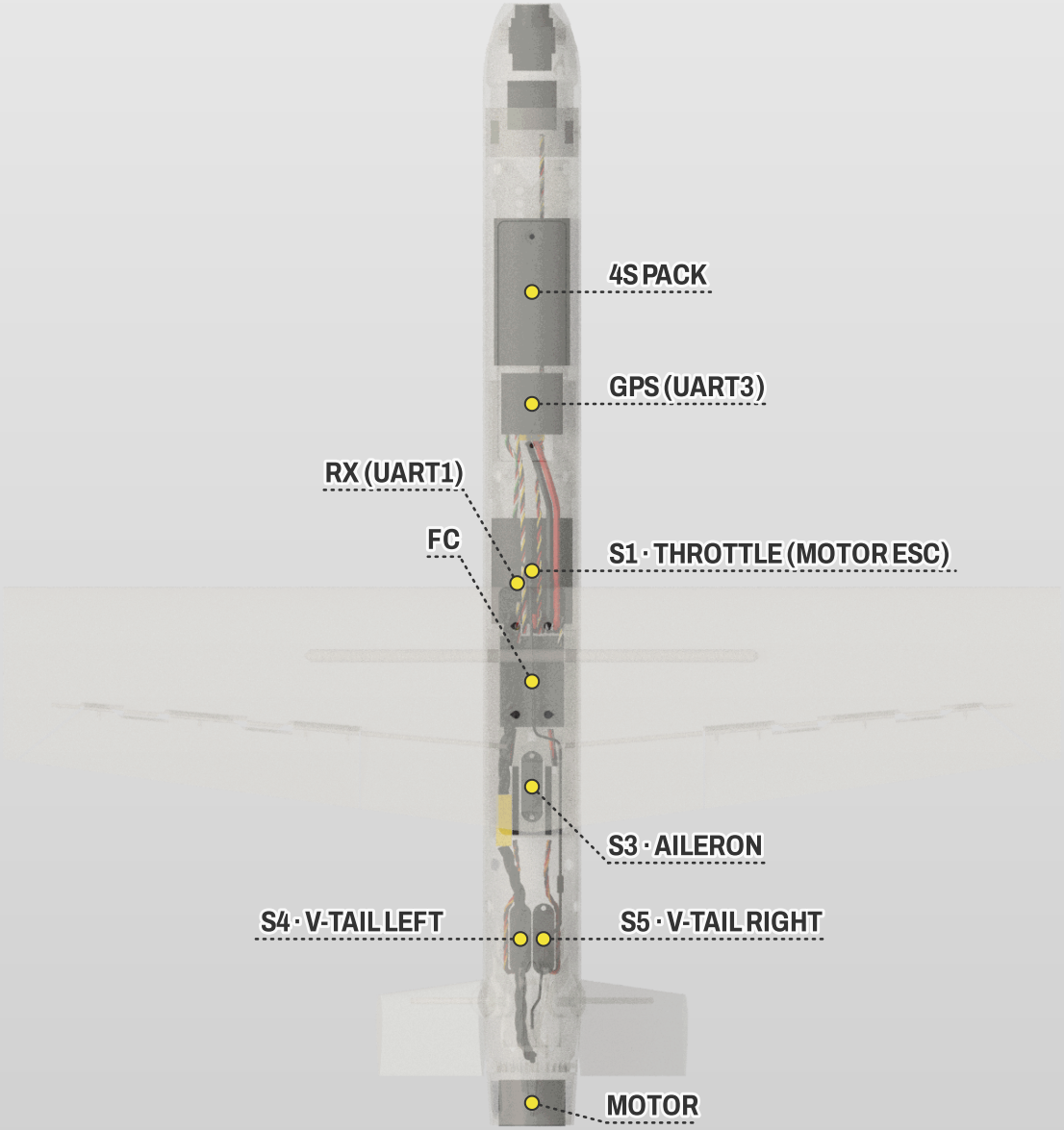
WIRING DIAGRAM

The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the next page has the diagram at full size with the receiver, GPS and video ports.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	ESC (DShot300, bidirectional)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)

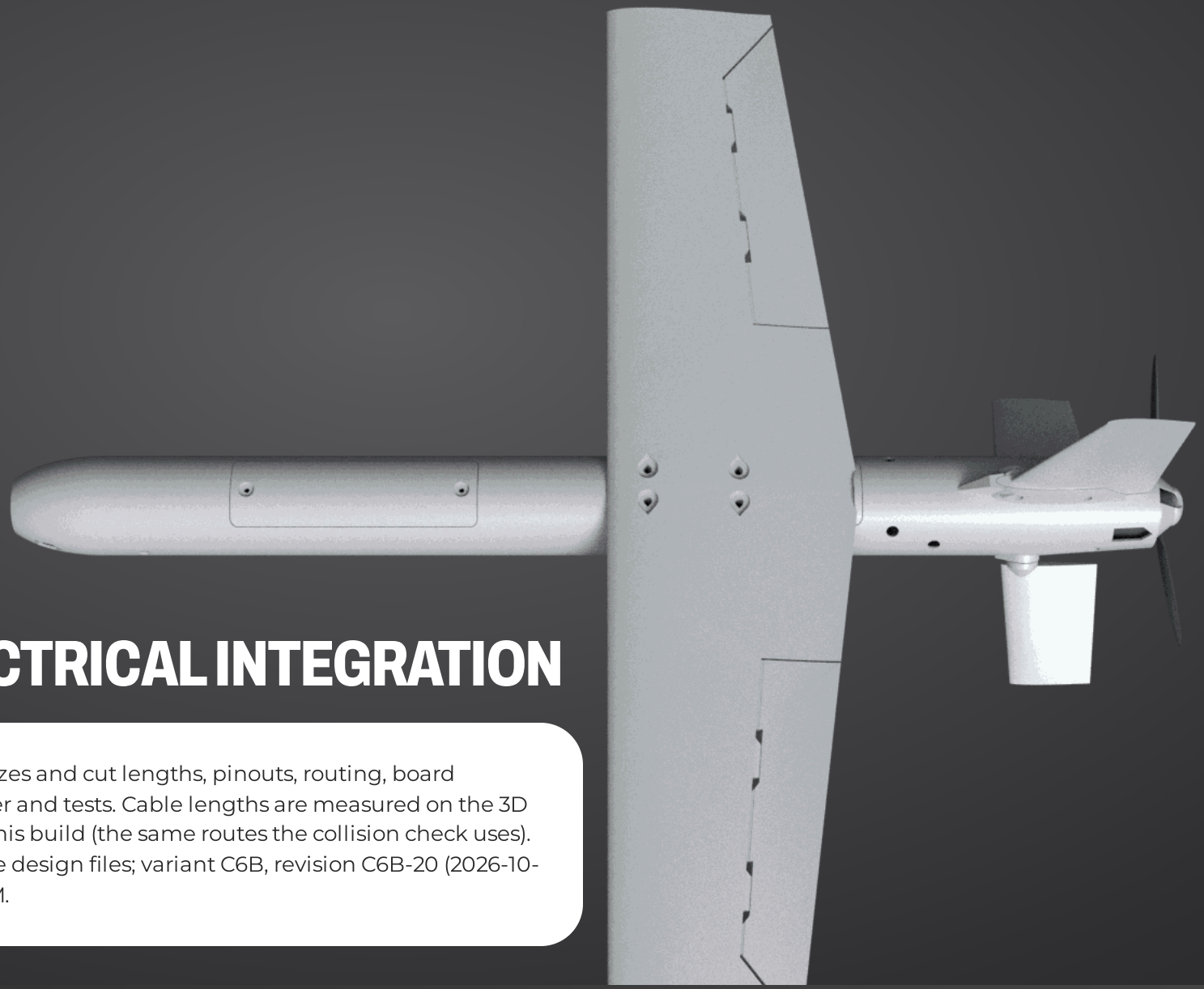


WIRING DIAGRAM (FULL SIZE)



PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	ESC (DShot300, bidirectional)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)

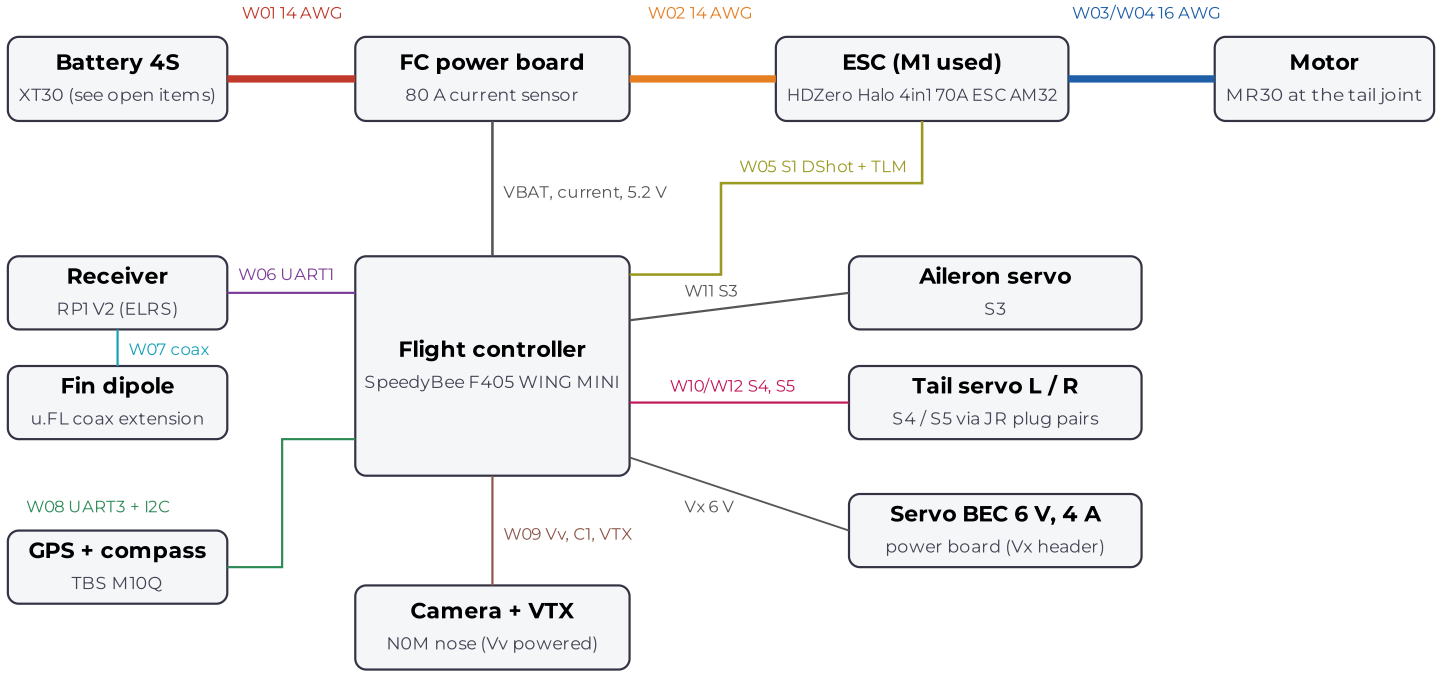
PORT	DEVICE	SETTING
UART1 (TX1/RX1) / SERIAL1	RadioMaster RP1 V2 ELRS receiver	SERIAL1_PROTOCOL = 23
UART3 (TX3/RX3) / SERIAL3	TBS M10Q GPS (compass on SDA/SCL)	SERIAL3_PROTOCOL = 5
UART5 (TX5/RX5) / SERIAL5	DJI O4 Air Unit (O4 noses only)	SERIAL5_PROTOCOL = 42 (O4 noses)
UART6 / SERIAL6	Wireless USB extender (ground setup)	SERIAL6_PROTOCOL = 2



ELECTRICAL INTEGRATION

Power path, wire sizes and cut lengths, pinouts, routing, board settings, build order and tests. Cable lengths are measured on the 3D harness model of this build (the same routes the collision check uses). Generated from the design files; variant C6B, revision C6B-20 (2026-10-16), flight nose N0M.

ELECTRICAL INTEGRATION



SAFETY

LiPo packs deliver over 100 A into a short. Work with the pack unplugged; first power-up through a smoke stopper or a current-limited supply; prop off for every bench test except the final static run; never charge inside the aircraft.

The battery feeds the flight controller's power board first, so its SpeedyBee 80 A current sensor measures everything; the power board feeds the ESC (HDZero Halo 4in1 70A ESC AM32, channel M1; M2 is the spare) and makes 5.2 V for the electronics and 6 V for the servos. The motor plugs in at the tail joint (MR30) so the tail module comes off. The receiver's antenna is a dipole inside the fin, clear of all metal.

ELECTRICAL INTEGRATION

Power budget and wire sizing

Cable	Circuit	Current A (pass / static)	Wire	Loop mm	R mOhm	Drop V	Loss W	dT K / pass	Connector rating
W01+W02	Battery to ESC (DC)	33 / 36	14 AWG	210	1.7	0.06 (0.4 %)	1.9	10	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	27 / 29	16 AWG	362	4.8	0.13 (0.9 %)	3.4	17	MR30: 30 A cont / 40 A for 1 min per contact

Currents from the propulsion model: 32.6 A at top speed (passes of <= 8 s), 35.9 A at full throttle on the ground. Phase RMS = sqrt(2/3) x DC. Temperature rise: copper only, no cooling, for one 8 s pass (an upper bound). Drop in % of the nominal 14.8 V.

Electronics: FC BEC 5.2 V / 2 A continuous (FC, GPS, receiver about 1.2 A together); servo BEC 6 V / 4 A continuous for three micro servos. Signal and servo wires: 26-28 AWG. Main DC path: 14 AWG (16 AWG would rise about 25 K per pass); motor phases: 16 AWG.

Cable schedule (cut list)

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under the GPS	FC power board BAT+ / BAT- pads	red BAT+, black BAT-	14	91	**120**	XT30 solder cups + 3/16 in heat shrink; pads tinned, 3 mm stripped	Over the ESC at z +2, down the 5 mm gap ahead of the FC stack.
W02	ESC feed	FC power board ESC+ / ESC- pads	HDZero Halo 4in1 70A ESC AM32 BAT+ / BAT- pads (XT60 lead removed)	red +, black -	14	14	**35**	Solder both ends; the ESC's low-ESR capacitor across its BAT pads (leads <= 10 mm)	Up the gap onto the ESC's rear end. 80 A current sensor in the path.
W03	Motor phases, ESC side	HDZero Halo 4in1 70A ESC AM32 M1 pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the tail joint	3 x black (mark A/B/C)	16	81	**110**	Solder to the M1 pads; MR30 solder cups + 3 mm heat shrink	Up the gap and over the FC's left half (servo header row on the right).

ELECTRICAL INTEGRATION

Cable schedule (cut list) (continued)

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat shrink)	3 x black	16	100	**140**	MR30 solder cups; butt splices with 3 mm heat shrink	Through the joint's upper quadrant, along the F3 top left of the fin dipole, down to the motor plate (M1). Unplug the MR30 before pulling the tail module.
W05	ESC signal	HDZero Halo 4in1 70A ESC AM32 8-pin SH1.0 lead (only M1, GND, TLM kept)	FC S1 header (signal, GND); TLM to the R4 pad	yellow M1 signal, black GND, white TLM	28	18	**45**	SH1.0 plug at the ESC (supplied lead, unused wires cut back and insulated); 1-pin Dupont housings at the FC header	Do NOT connect the lead's VBAT or CURR wires; never feed ESC voltage to the servo rail (Vx).
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the ESC	FC receiver header (5V, GND, RX1, TX1)	red 5V, black GND, yellow RP1-TX -> RX1, white RP1-RX -> TX1	28	22	**45**	Solder at the receiver; 4-pin Dupont housing at the FC	Low over the ESC, up the gap at y -5.5.
W07	ELRS antenna coax	RP1 V2 u.FL	Dipole feed at the centre of the fin dipole (z 39.5)	1.13 mm coax	coax	241	**275**	Two pieces joined by a u.FL plug/jack 4 mm ahead of the tail joint: the receiver-side extension (u.FL male to u.FL female) and the T-antenna's own coax, which runs up the fin's channel beside the lower leg to its feed	Under the wing nut bars, right of the aileron hardware, over the tail servos, up the fin channel. Unplug the u.FL joint to remove the tail module. Antenna side (joint to feed) 92 mm: buy a 2.4 GHz T/dipole antenna with at least 110 mm of coax (the RP1's stock T-antenna is shorter), receiver side 148 mm.
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC GPS interface with compass (4V5, GND, RX3, TX3, SDA, SCL)	6 x 28 AWG (the model shows the bundle as 4)	28	90	**130**	Supplied GPS plug; solder or GH plug at the FC	Under the GPS carrier; leave a slack loop so the hatch lifts off.

ELECTRICAL INTEGRATION

Cable schedule (cut list) (continued)

ID	Cable	From	To	Conductors	AWG	Route mm (model)	Cut mm	Terminations	Route / notes
W09	Video trunk (N0M nose)	Nose joint plug (camera on the nose; VTX on the nose or, dome nose, under the hatch)	FC pads: Vv, G, C1 (camera in), VTX (video out)	red +, black GND, yellow camera video, white VTX video	28	211	**275**	4-pin JST plug pair at the nose joint; solder at the FC	Along the tube floor under the equipment tray, up through the tray's floor hole (x 182-194) to the FC.
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	brown GND, red +6 V, orange signal	26	64	**stock**	Stock servo plug	Along the floor and through the tail joint; the plug pair separates the tail module. Needs a stock lead of at least the route length + 30 mm.
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	brown, red, orange	26	15	**stock**	100 mm stock extension (BOM line 35)	Along the floor to the FC's rear edge.
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	brown, red, orange	26	60	**stock**	Stock plug	Under the floor beside the servo, up behind the FC.

Cut length = route length on the model x 1.10 (slack, 1.15 for plugged leads) + the termination allowances, rounded up to 5 mm. Total cut length of made cables: 1175 mm. Stock servo leads are used as supplied (W10 servo side, W11).

Connections and pinouts: SpeedyBee F405 WING MINI

Pad / header	Connects	Setting / note
BAT+ / BAT- (power board)	W01 battery lead	14 AWG red / black
ESC+ / ESC- (power board)	W02 ESC feed	14 AWG red / black
S1 header: signal, GND	W05 ESC M1 signal + GND (no + wire)	DShot300 bidirectional (SERVO1_FUNCTION 70)

ELECTRICAL INTEGRATION

Connections and pinouts: SpeedyBee F405 WING MINI (continued)

Pad / header	Connects	Setting / note
R4 pad	W05 ESC TLM (optional)	AM32 serial telemetry, fallback rpm source
S3 header	W11 aileron servo	SERVO3_FUNCTION 4
S4 / S5 headers	W10 tail servos L / R (via extensions)	SERVO4/5_FUNCTION 79 / 80 (V-tail)
Receiver header: 5V, GND, RX1, TX1	W06 RP1 V2	CRSF, SERIAL1_PROTOCOL 23
GPS interface: 4V5, GND, RX3, TX3, SDA, SCL	W08 TBS M10Q	SERIAL3_PROTOCOL 5; compass on I2C
Vv, G, C1	W09 camera (power + video in)	Vv = battery voltage by default: the camera must accept 4S
Vv, G, VTX (T5 for HD)	W09 VTX (power + video out)	Analog OSD on the FC; HD/DJI: T5/R5 MSP DisplayPort
USB extender (UART6)	Wireless USB extender	Setup on the ground; not flown

HDZero Halo 4in1 70A ESC AM32: 8-pin SH1.0 lead

Pin	Wire	Use
M1 signal	yellow	W05 -> FC S1 signal
GND	black	W05 -> FC S1 GND
TLM	white	W05 -> FC R4 (optional)
VBAT	-	Cut back and insulate: the FC is powered by its own power board
CURR	-	Cut back and insulate: the FC's own 80 A sensor is used
M2, M3, M4	-	Cut back and insulate (unused channels; M2 is the spare)

> Never connect the ESC lead's VBAT pin to the FC's servo header (+) or any 5 V pad: it is battery voltage.

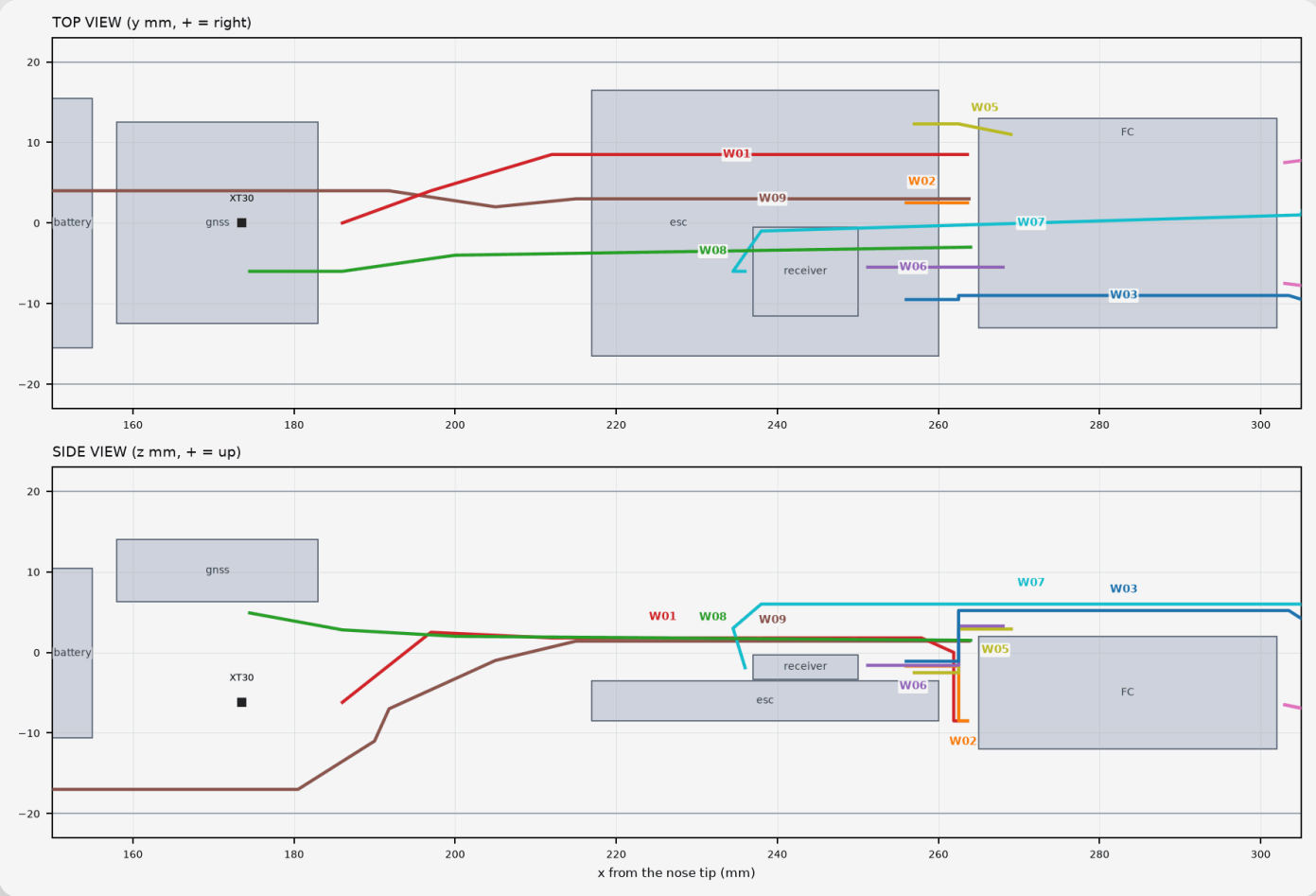
ELECTRICAL INTEGRATION

Connectors

Connector	Gender / pins	Note
XT30 (battery)	Female on the aircraft lead, male on the pack	Red +, black -; check polarity on the housing's chamfer
MR30 (motor)	Female on the ESC side, male on the motor side	Phase order free: set the direction in AM32, not by swapping wires
JR servo plug	Brown GND, red +6 V, orange signal	Same orientation on every header: signal towards the board edge marked S
u.FL (ELRS coax)	Plug on the receiver and at the tail joint	Max about 30 mating cycles; tape the joint, no tension on the coax

ELECTRICAL INTEGRATION

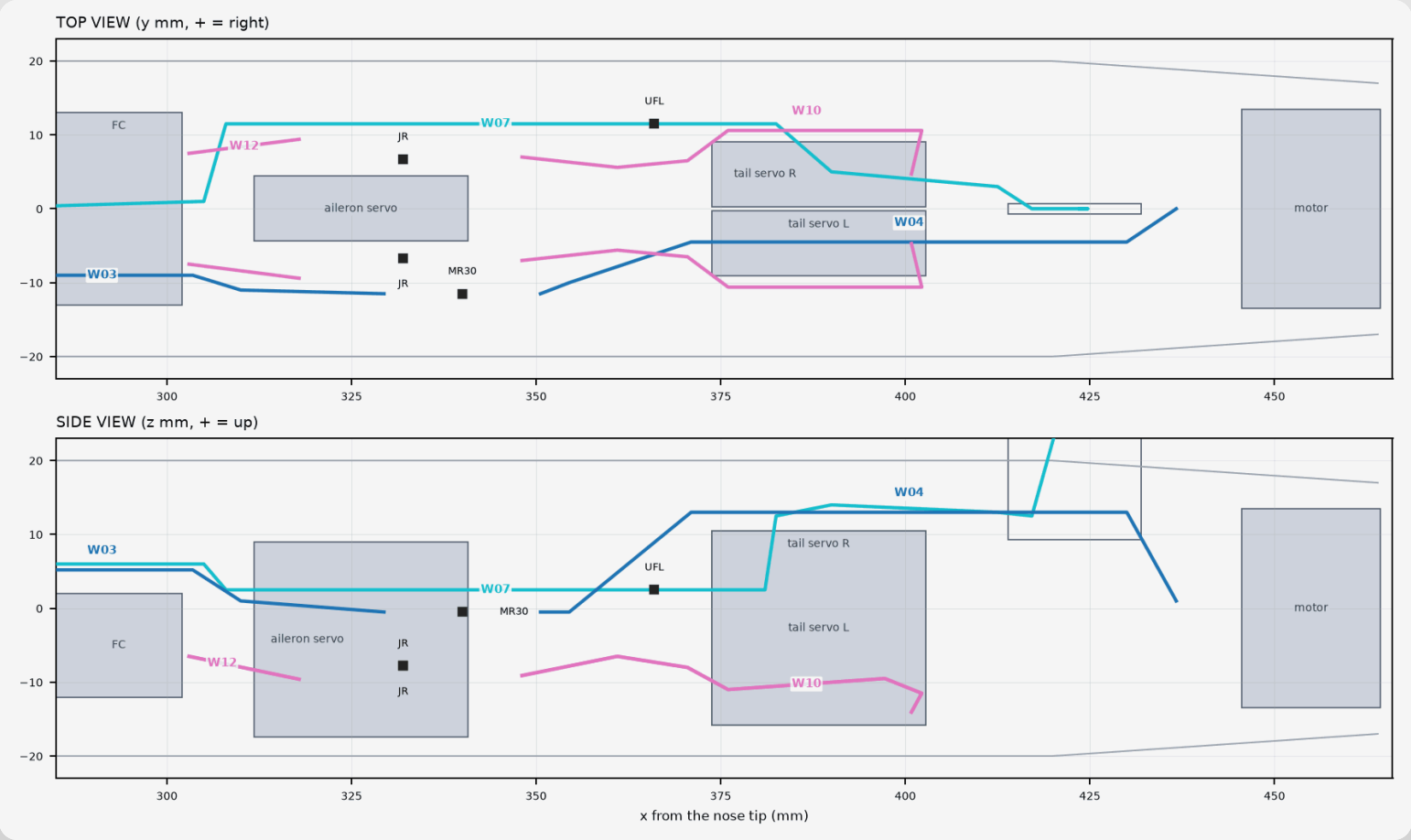
Routing



Front: battery bay, wing bay (ESC, FC, receiver).

ELECTRICAL INTEGRATION

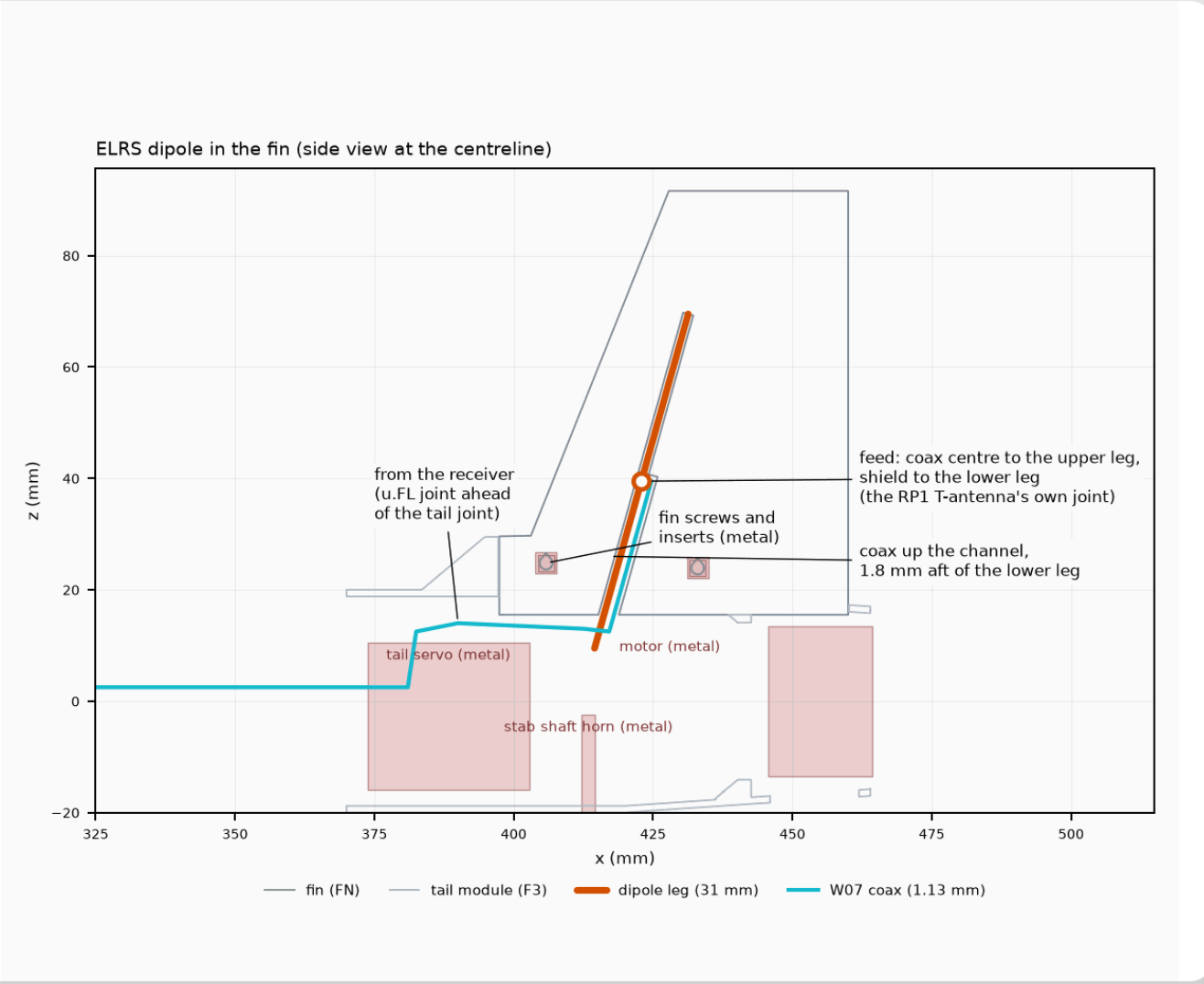
Routing



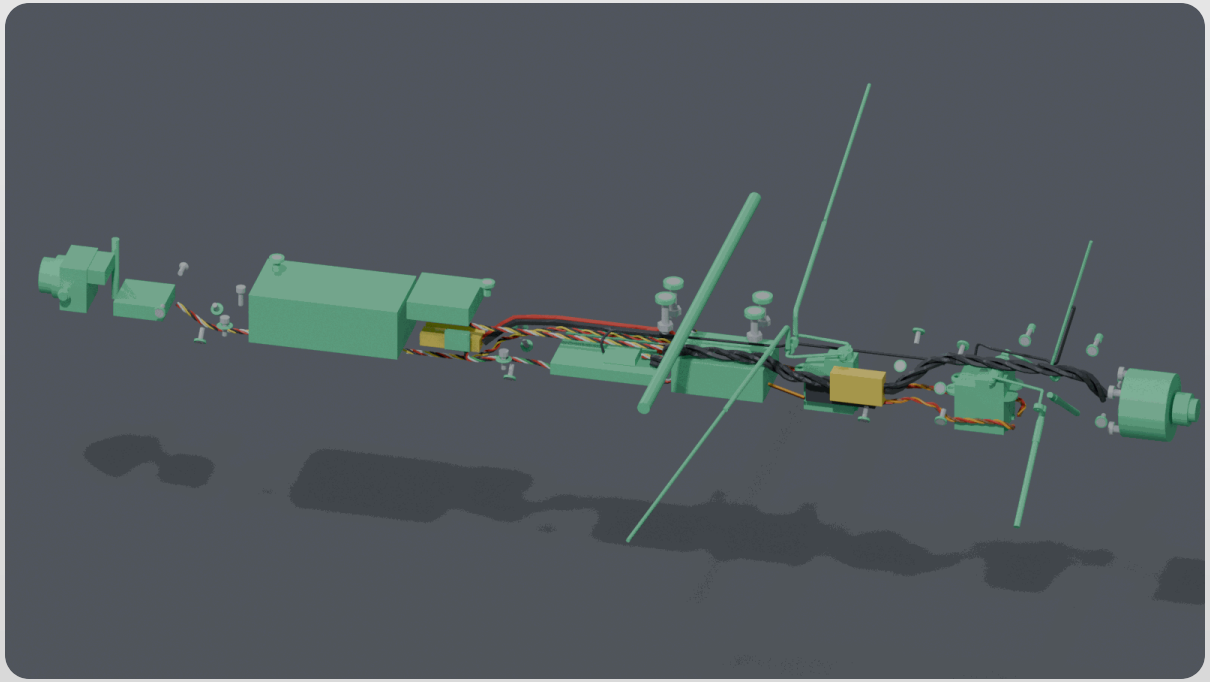
Rear: aileron servo, tail joint, tail module, fin. Cable routes from the 3D harness model (x from the nose tip); squares are connector pairs. Every route is collision-checked against every printed part, component and cable.

ELECTRICAL: RECEIVER ANTENNA

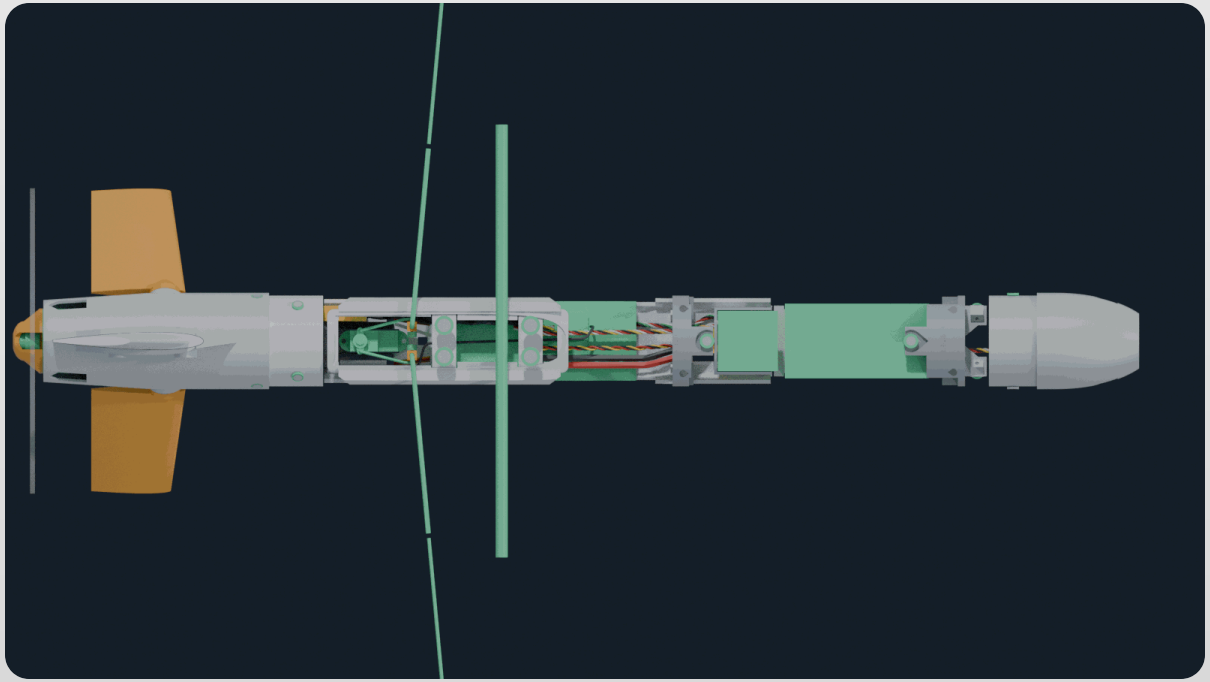
The RP1 V2 T-antenna straightened into a vertical dipole: two 31 mm legs in a closed channel inside the fin on its 30 % chord line, the feed (the T's joint) at the centre. Its own coax runs down a parallel channel beside the lower leg, out of the fin root through the slot in the shoe floor into the tail module, and meets the 1.13 mm extension at the u.FL joint ahead of the tail joint. Clearance to the nearest metal (fin screw): 10.9 mm. Fit: thread the antenna and coax up into the channels from the root before the fin goes into its shoe; no glue, a strip of tape round the coax at the root face.



ELECTRICAL: HARNESS



Render: harness with the fuselage hidden.



Render: top view, fuselage cut open.

Lanes in the 5 mm gap between the ESC and the FC stack (left to right, y in mm): motor phases W03 -9.5, receiver W06 -5.5, ESC feed W02 +2.5, battery W01 +8.5, ESC signal W05 +12.3. The video trunk W09 and the GPS lead W08 arrive over the ESC at z 0.5-1.5; the receiver coax W07 runs higher (z +6).

ELECTRICAL: HARNESS



Render with the fin and tail module see-through: the dipole (green) and the antenna coax (cyan) entering the tail module from the u.FL joint, along the top and up the fin's channel beside the lower leg to the feed.

ELECTRICAL INTEGRATION

Board and ESC settings

Item	Setting	Why
Power board: servo BEC voltage jumper	6 V	Emax ES9052MD: 1.3 kgf cm at 6 V (1.1 at 5 V). Vx header = 6 V.
Power board: servo BEC supply jumper	Servo BEC (default)	Keep: the ESC has no BEC; never power Vx from the ESC lead.
Power board: Vv (VTX/camera) jumper	Battery voltage (default)	Only if camera and VTX accept 4S (13-17 V); otherwise 5 V (shares the servo BEC; then 6 V when the servo jumper is 6 V).
ESC (AM32): protocol	DShot300, bidirectional on	SERVO_BLH_MASK 1, SERVO_BLH_BDMASK 1, SERVO_BLH_OTYPE 5.
ESC (AM32): motor direction	Set on the bench	Pusher prop: the motor turns the prop so that it blows aft. Reverse in AM32 if wrong.
ESC (AM32): brake on stop	On	Holds the prop still for the belly landing.
ESC (AM32): motor poles	14	2207 motors are 12N14P: SERVO_BLH_POLES 14 for correct rpm.
FC: battery monitor	BATT_MONITOR 4, BATT_VOLT_MULT 11.0, BATT_AMP_PERVLT 50	Calibrate both against a meter (see the tests).
FC: throttle limit	THR_MAX 100 (no cap)	No cap needed: this motor and pack cannot overspeed the prop. Full-throttle bench runs <= 10 s (MR30, XT30 heating).

Build order

#	Step
1	Bench, before anything goes into the fuselage: flash ArduPilot (SpeedyBeeF405WING), load the parameter file, set the power-board jumpers (table above).
2	ESC: unsolder the supplied XT60 lead. Tin the BAT pads and solder W02 (14 AWG, cut length in the schedule) and the low-ESR capacitor across them (short leads, capacitor laid flat beside the receiver position). Solder W03 to the M1 pads. Cut the SH1.0 lead's VBAT, CURR, M2-M4 wires back to 5 mm and insulate each.
3	FC: solder W01 and the free end of W02 to the BAT and ESC pads (14 AWG: 3 mm strip, pre-tin, 2-3 s per joint with a 60-80 W iron). Solder W08 (GPS), W09 (video) and the TLM wire of W05 (R4). Fit the 1-pin Dupont housings of W05 on S1 (signal + GND) and the 4-pin housing of W06 on the receiver header.
4	Continuity and insulation, unpowered: BAT+ to BAT- > 1 kOhm (capacitor charging is normal), no short between any motor phase and BAT-, polarity of the XT30 matches the pack.

ELECTRICAL INTEGRATION

Build order (continued)

#	Step
5	Smoke stopper first power-up (a current-limited supply, 16 V / 0.5 A, or a smoke stopper on the pack): FC boots, ESC beeps, no heat anywhere. Then the pack directly.
6	On the equipment tray's wing-bay floor, front to back: ESC on its foam tape (component side up), receiver on the ESC (left of centre), FC stack 5 mm behind the ESC (arrow forward, servo header row on the right). Route W01, W02, W05 and W06 down/up the 5 mm gap in their lanes (W01 at y +8.5, W02 at y +2.5, W05 at y +12.3, W06 at y -5.5).
7	Motor: W03 up the gap at y -9.5 and over the FC's left half to the MR30 low on the left ahead of the tail joint; W04 from the MR30 through the joint's upper quadrant and along the F3 top 4.5 mm left of centre to the motor.
8	Receiver antenna: W07 from the RP1 up over the bay at z +6 (under both wing nut bars), to the right side behind the FC, through the tail joint (u.FL joint here, taped), up over the tail servos to the fin dipole. The dipole: RP1 T-antenna straightened, 31 mm legs, threaded up into the fin's closed channel (30 % chord line) from the root before the fin goes into its shoe; no glue, tape round the coax at the root face.
9	Servos: aileron lead W11 under the floor to S3; tail servo leads through the joint to the JR plug pairs lying flat beside the aileron servo, extensions to S4 (left) and S5 (right).
10	Strain relief: Kapton tape at every lane change, nothing resting on the motor phases (W03/W04) or the ESC's FETs, a slack loop on W08 for the hatch, MR30 and the u.FL joint reachable through the tail joint.

Power-up and tests

Test	Procedure / pass criterion
Voltage	Read the pack voltage with a meter and in ArduPilot; adjust BATT_VOLT_MULT until they agree within 0.05 V.
Current	Run the motor (prop off) at a steady setting with a clamp meter or a watt meter on the pack lead; adjust BATT_AMP_PERVLT.
Motor direction and rpm	Prop off: short burst, check direction (pusher) and that ESC rpm appears in ArduPilot (bidirectional DShot).
Servos	Direction and travel of each surface against the ArduPilot document; no binding at full deflection; aileron reflex 3 deg TE up at neutral.
Radio and failsafe	Bind the RP1 V2; switch the transmitter off: RTL after 5 s. Range check at reduced power with the antenna in the fin.
GPS and compass	3D fix outdoors; compass calibration with the ESC and battery installed (current drawn near the compass).
Thermal	Static full throttle (prop on, aircraft held) for at most 10 s: ESC, wires, XT30 and MR30 hand-warm afterwards (connectors are near their ratings).

ELECTRICAL INTEGRATION

Power-up and tests (continued)

Test	Procedure / pass criterion
Video	Camera image and OSD (voltage, current, ground speed) in the goggles; VTX power and band legal for the event.

Open items (decide before ordering or flying)

#	Open item
1	XT30 rating: Amass XT30 is rated 15 A continuous / 30 A peak; this aircraft draws 33 A on a pass and 36 A at static full throttle. FPV builds often run XT30 this hard for seconds, but it is outside the rating. XT30 kept (owner 2026-10-11): ready-made 4S 650 packs come with XT30 only; full power in passes of 10 s or less, then at least 30 s at low throttle.
2	Halo ESC thickness: modelled 5 mm with components. Measure on receipt; in the tube there are about 13 mm above it to the wing-window filler, so a thicker board fits: only the receiver and the leads on its top rise with it.
3	Halo 8-pin lead: the pin order is printed on the ESC; confirm M1, GND and TLM on the board before cutting the other wires.
4	FC power-board pad positions: the routes end at the stack's front face; the cut lengths include 25-30 mm for reaching the pads wherever they sit on the board.

Tools and consumables

#	Item
1	Temperature-controlled iron 60-80 W with a chisel tip (14 AWG pads), fine tip for 28 AWG; leaded solder 0.5-0.8 mm; no-clean flux.
2	Silicone wire: 14 AWG red + black, 16 AWG black (3 phases), 28 AWG in four colours; 1.13 mm u.FL extension.
3	Heat shrink 1.5, 3, 4.8 mm; Kapton tape; double-sided foam tape; XT30 pair, MR30 pair, 1-pin and 4-pin Dupont housings with crimps.
4	Multimeter, smoke stopper or current-limited bench supply, clamp or watt meter for the current calibration.

ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 90B on the SpeedyBee F405 WING MINI (firmware target SpeedyBeeF405WING, ArduPlane 4.5 or later). Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), then work through the bench checks.

Flight controller: on the wing-bay floor, arrow forward; servo BEC jumper set to 6 V for the ES9052MD servos.

Firmware: ArduPlane 4.5 or later · **Board:** SpeedyBee F405 WING MINI (ArduPilot target: SpeedyBeeF405WING)

The setup document (PDF) and the starting .param file are in the downloads. Verify every value on the bench before flight.



FLIGHT TEST AND SPEED STAGES

The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	STAGE throttle limit	Expected level speed
30 m/s	33 %	24 m/s
35 m/s	37 %	28 m/s
40 m/s	42 %	32 m/s
45 m/s	47 %	37 m/s
50 m/s	53 %	42 m/s
55 m/s	58 %	46 m/s
60 m/s	63 %	50 m/s
65 m/s	68 %	53 m/s
70 m/s	74 %	58 m/s
75 m/s	79 %	62 m/s
80 m/s	85 %	66 m/s
85 m/s	91 %	71 m/s
90 m/s	96 %	74 m/s

SKYG

SKYG SPEED 90B

